



Noninvasive Treatment of Diabetes

An In-Vitro Analysis of Chitosan-Insulin Nanoparticles for Oral Insulin Delivery

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Introduction

Diabetes mellitus, a metabolic disease of digestive system, affects more than 100 million people around the world and appears mostly in two types: Type1 and Type2. This happens when the pancreas produces little to no insulin or when the body cells cannot use the insulin properly. **Insulin is a protein hormone produced in the β -cells of pancreas.** It regulates blood sugar levels by allowing the body to use glucose from carbohydrates in the food eaten for energy, or to store it for future use.

To maintain the blood glucose levels, Insulin is usually administered by the subcutaneous route as injection. However, approx. 60% patients fail to achieve long-term glycemic control due to poor patient compliance owing to the use of needles and the complexity of the insulin treatment. To overcome such problems, different routes of insulin administration are considered. The oral Insulin delivery is the preferred route, but it remains unsuccessful due to large particle size, hydrophilicity and instability within the GI tract resulting rapid enzymatic degradation and poor intestinal absorption.

Nanotechnology can help. Chitosan is a natural polymer derived from chitin by partial deacetylation. It is biodegradable, biocompatible, non-toxic and non-immunogenic polymer. This project aims to make a self-assembly of Chitosan and Insulin nanoparticles without the use of any harsh chemicals by encapsulating insulin into chitosan for the use in oral insulin delivery. It also aims to observe the

Objectives

- Make a self-assembly of Chitosan and Insulin nanoparticles by encapsulating insulin into chitosan without using any harsh chemicals.
- Determination of the physicochemical properties of the Chitosan-Insulin nanoparticles.
- In-Vitro analysis of the nanoparticles for Insulin encapsulation efficiency and Insulin release profile in Human digestive system pH conditions.

Question and Hypothesis

Question: How do different weight ratios of Chitosan-Insulin nanoparticles improve insulin retention and release at different pH corresponding to the human digestive system GI tract?

Hypothesis: If 3 samples of different weight ratios (1:1, 2:1 and 3:1) of Chitosan and Insulin are taken in the self-assembled Chitosan-Insulin nanoparticle, their particle sizes are analyzed and they are put in the pH conditions corresponding to the human digestive system, then the Chitosan-Insulin weight ratio 1:1 will be able to achieve the lowest particle size and all the weight ratios of the self-assembled nanoparticles will be able to achieve more concentration of released Insulin than the control sample of only insulin without any Chitosan nanoparticle carrier.

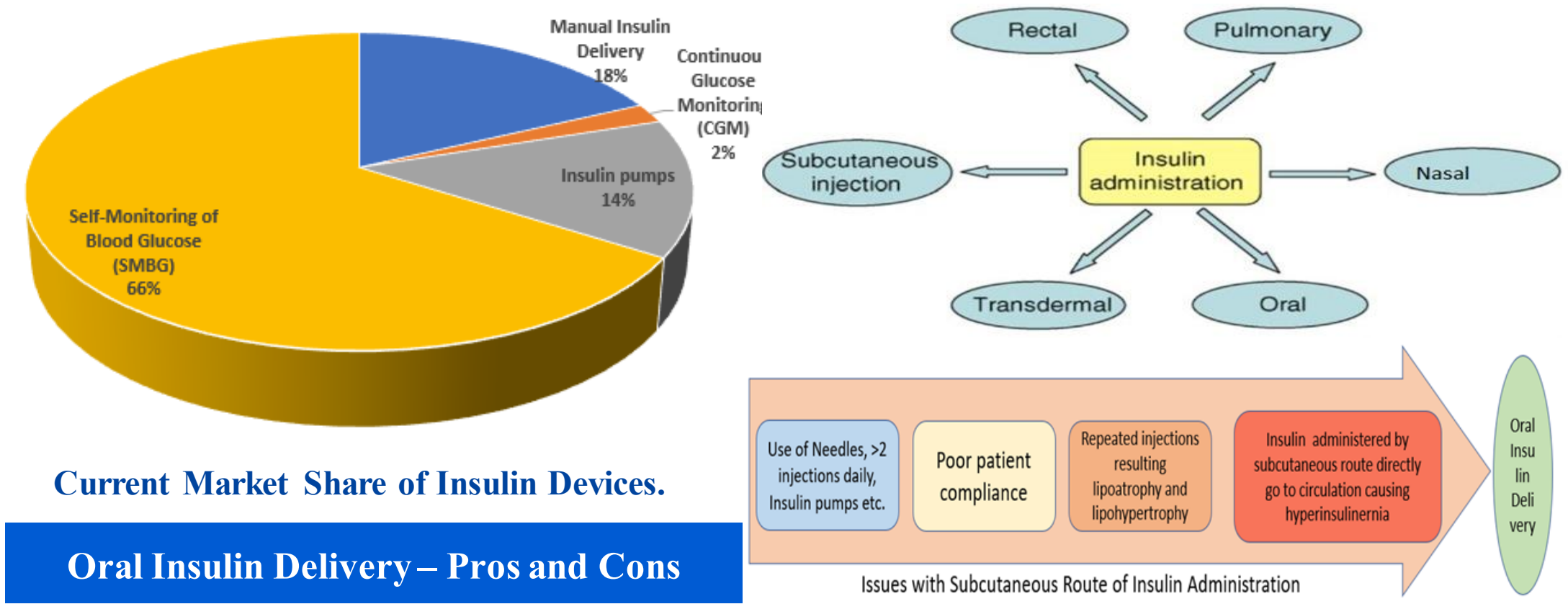
Independent Variables: 3 different weight ratios of Chitosan - Insulin nanoparticles.
Dependent Variables: Particle Size, Zeta potential and Concentration of the released Insulin.
Constants: The pH conditions, Volume of the samples, Degree of deacetylation of the samples, Temperature, etc.

Background Research

Type 1 and Type 2 Diabetes and Current Treatments

- Diabetes mellitus is a metabolic disease characterized by high blood sugar (glucose) levels due to defects in insulin secretion or the body's ability to use insulin.
- In Type 1 diabetes (5-10% of diabetes patients) the pancreas doesn't produce any insulin at all due to damage in the pancreas β -cells.
 - Type 2 diabetes is more common (90-95% of diabetes patients) and it occurs due to a low production of insulin or the body does not effectively use the produced insulin.
 - There are other types of diabetes such as gestational diabetes that occurs from pregnancies. However, they are not as common as Type 1 and type 2 diabetes.

Insulin is a protein hormone secreted by pancreatic β -cells, it plays a crucial role in glucose metabolism regulation and is used as a drug to control Type-1 and Type-2 Diabetes Mellitus. It is administered by the subcutaneous route as injection/Insulin pumps etc. which significantly reduces morbidity and mortality.



Application of Biotechnology - Nanotechnology to solve the problem

- Recent advances in Nanotechnology have made the mass production of therapeutic micro-molecules, such as proteins, nucleic acids and polysaccharides, feasible.
- The nanoparticles can help in drug delivery due to its versatility in targeting tissues, enabling deep molecular targets as well as sustained drug release.
- Nanoparticles display excellent efficiency as potent proteins, peptide, vaccines and DNA carrier.
- Thus, nanoparticles offer a better way for delivering the hydrophilic drugs such as insulin. Several natural polymers like, alginate, pectin, gelatin, chitosan, etc. can be used to prepare the nanoparticles.
- Chitosan is a natural bio-polymer derived from chitin by partial deacetylation. Chitin is extracted from crustacean (crabs, shrimps and lobsters) shells, fungi or yeast. It is a biodegradable, biocompatible, non-toxic and non-immunogenic muco-adhesive polymer.
- This project aims to prepare a self-assembly of Chitosan-Insulin nanoparticles for oral insulin delivery.

Materials and Methods

Materials

- Glacial Acetic Acid. 60.052 g/mol 1.049 g/cm³
- Tris (hydroxymethyl) Aminomethane 500 g
- Chitosan 70-150 mPa.s
- Insulin 27 USP units/mg
- Distilled Water 10 L
- Erlenmeyer flasks 500 ml - 8 Nos
- Graduated cylinder 100 ml - 4 Nos
- Glass funnels and filter stand with Burette clamp.
- Test tubes - 20 nos
- Hydrochloric Acid - 30 ml
- Millipore 0.5 μ m filter
- Sodium Hydroxide (NaOH) solution -100 ml
- Sodium Nitrite (NaNO₂) - 50 gm
- Portable pH meter.
- Low Profile Vortex Mixer.
- UV-vis Spectrophotometer.
- Infrared Spectroscope.
- Zetasizer Nano ZS.
- Glass Capillary Viscometer.



Chitosan Nanoparticle Preparation

- Main aspects of Chitosan Nanoparticles:
- Two main aspects of chitosan for the suitability of chitosan for drug delivery. Molecular Weight (MW) and Degree of deacetylation (DD) affecting the DNA condensation and release, immune-adjutant activity and cellular uptake.
- Purification of Chitosan:
- Chitosan is purified by oxidative degradation with Sodium Nitrate NaNO₂ at room temperature.
- Make 1% acetic acid solution by adding 2 ml 99.8% Glacial acetic acid in 198 ml distilled water.
 - Weight 2 gm of dry chitosan and pour 200 ml of the 1% acetic acid solution in it.
 - Stir the solution to dissolve the chitosan to make 1%(w/v) chitosan dissolved in 1% acetic acid.
 - Add 10 ml of NaNO₂ in 1% acetic acid drop wise for 30 min and stir it vigorously. Keep it for 3 h.
 - Neutralize the reaction mixture with 1N NaOH to pH 7.4 to complete precipitation of chitosan.
 - Wash the precipitate with distilled water, and dry it overnight at 55°C to obtain purified chitosan.

Determination of Molecular weight and Degree of deacetylation

- Determination of the Molecular weight by Glass Capillary Viscometer:
- Stabilize temperature of the water bath at 25°C and monitor for 10 min before the first measurement.
 - Filter the 1% chitosan solution, twice through a 0.5 μ m filter, and Place 10 ml of the filtered chitosan solution in the viscometer.
 - Add 10 ml of the filtered solvent was to the viscometer and measure the flow time of the resulting diluted solution 5 times.
 - Dilute the solution by adding 10 ml solvent each time for 5 times. Due to the pressure difference at the top and bottom of the viscometer, the flow time of the solution in the capillary type is introduced in the calculations.
- Determination of the Degree of deacetylation (DD) by titration:
- Dissolve 1 g of chitosan in 250 ml of 0.1 mol/L HCl solution.
 - Add 100 ml distilled water and calculated amount of NaCl to adjust the ionic strength to 0.1 mol/L.
 - Perform titration by adding the titrant solution of 0.1 mol/L NaOH to the chitosan solution gradually.
 - Record the volume of NaOH added and pH values using digital pH-meter.

DD (%) of chitosan is calculated by using the following equation:

$$DD (\%) = \frac{203.2}{42.0 + \frac{1000m}{C_{NaOH} (V_2 - V_1)}} \times 100$$

- m is the amount of chitosan (g) in the solution. C_{NaOH} is the concentration of NaOH solution (mol/L).
- V₂ – V₁ is the volume of NaOH solution consumed in titration of amino groups of chitosan.
- 203.2 is the molecular weight of the acetylated monomeric unit of the polysaccharide.
- 42.0 is the MW difference between the deacetylated monomeric unit and acetylated monomeric unit.
- 1000 is the conversion factor of milliliters to liters and 100 is the conversion factor of DD to percent.

Results and Interpretation

Molecular weight and Degree of Deacetylation

Sample No.	Molecular Weight (kDa)	Degree of Deacetylation (%)	CS:INS Weight Ratio
1	222.0	80.0	1:1
2	222.0	80.0	2:1
3	222.0	80.0	3:1

Table-1. Different samples of chitosan (CS) -insulin (INS) for self-assembled nanoparticles

- The data analysis consists of 3 main parts:
- Determination of the physicochemical properties, Insulin encapsulation efficiency and In-Vitro Insulin release profile of the nanoparticles.

Zeta-Potential of Insulin control solution and self-assembled Chitosan(CS)-Insulin(INS) Nanoparticles

Trials	Control Solution (Insulin)	Sample-1 (CS:INS 1:1)	Sample-2 (CS:INS 2:1)	Sample-3 (CS:INS 3:1)
1	-10.5	2.8	10.6	20.5
2	-8.6	3.6	14.3	24.3
3	-6.9	4.0	11.2	18.5
4	-12.1	3.1	15.0	19.6
5	-10.9	2.5	10.9	22.7
Mean	-9.8	3.2	12.4	21.1

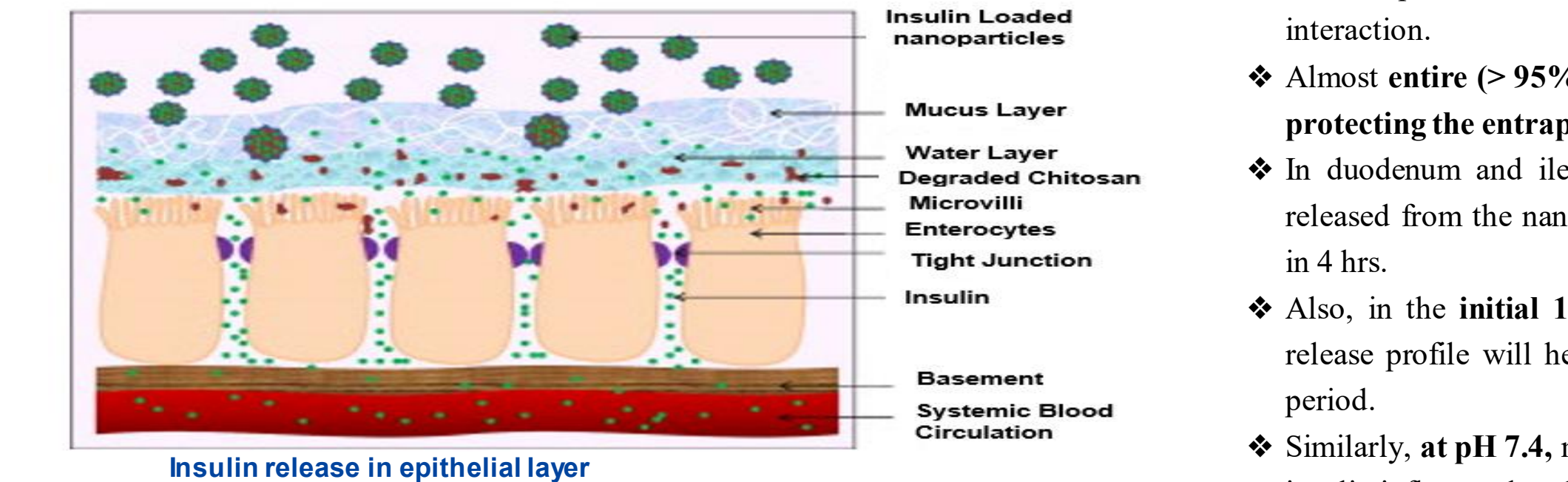
Table-3. Zeta Potential of Chitosan(CS)-Insulin(INS) nanoparticles (In mV)

- The mean particle size of the Chitosan-insulin weight ratio 1:1 is 210.2 nm. For 2:1, 463.6 nm, and for 3:1, 531.3nm. the nanoparticles formed at 1:1 weight ratio achieved the lowest particle size and being the smallest, it is most preferred for delivering insulin orally.
- The mean zeta potential of the Insulin control solution is -9.8 mV. However, the zeta potential of chitosan-insulin nanoparticles are subsequently higher at +3.2 mV, +12.4 mV and +21.1 mV positive charges of the chitosan molecules neutralized all the negative charges.

Trials	Sample-1 (CS:INS 1:1)	Sample-2 (CS:INS 2:1)	Sample-3 (CS:INS 3:1)
1	75.6	82.5	88.5
2	80.2	88.7	90.4
3	82.5	84.2	94.5
4	78.3	80.7	92.6
5	79.4	86.5	95.3
Mean	79.2	84.5	92.3

Table-4. Insulin encapsulation (%) in chitosan-insulin nanoparticles.

- EE (%) = (Ti-Fi)/Ti $\times$ 100
- Ti = Total amount of insulin. Fi = Free insulin.



Preparation of self-assembled chitosan-insulin nanoparticles

- Mix equal volume of both chitosan and insulin solutions and with a Vortex mixer, vortex for 30 sec.
- Dissolve the dry purified chitosan made before in 1 % acetic acid solution and stir for 30 min.
- Adjust the pH of the solution at 5.5 -5.7 by the addition of sodium hydroxide solution.
- Dissolve lab-grade Insulin (27 USP/ml) in 0.1M HCl solution and adjust the pH of the Insulin solution to 8.4.
- Incubate the mixture at room temperature for 30 min to completely form the self-assembled complexes.



Determination of physicochemical properties of the self-assembled nanoparticles

- Fourier transform infrared (FT-IR) spectroscopy analysis:
- Fourier transform infrared (FTIR) analysis is done using ATR-FTIR spectrometer using the Chitosan solution, Insulin solution and Chitosan-Insulin nanoparticle solution, and scan separately from 4000 to 500 cm⁻¹ at room temperature.
- Determination of Particle size and Zeta potential:
- The particle sizes and distributions were determined by Dynamic light Scattering (DLS) scan.



Insulin encapsulation efficiency of the nanoparticles

- Three different weight ratios of chitosan and insulin were mixed separately to obtain the 3 samples.
- The sample solutions were then centrifuged at 250 s⁻¹ for 30 min at room temperature. The clear supernatant was analyzed for insulin content using UV-vis spectrophotometer at 280 nm

The encapsulation efficiency (EE) is calculated by the formula: EE (%) = (Ti-Fi)/Ti $\times$ 100

Where EE (%) is the Encapsulation efficiency in percentage,



Measurement of the In-vitro insulin release profile

- Different pH levels corresponding to the human digestive system were simulated for the In-Vitro analysis.
- Prepare 3 buffer solutions with 3 different pH levels (pH 1.2, pH 6.8 and pH 7.4).
- The pH 1.2: pH of the stomach, pH 6.8: pH of Duodenum and pH 7.4: pH level of the Ileum Intestine.
- Immerse the Chitosan-insulin nanoparticles in the 3 buffer solutions with mild agitation and draw sample at 30 min interval.
- Measure the concentration of the released insulin sample drawn using UV spectrophotometer at 280nm.

Particle Size of the Self-assembled Nanoparticles

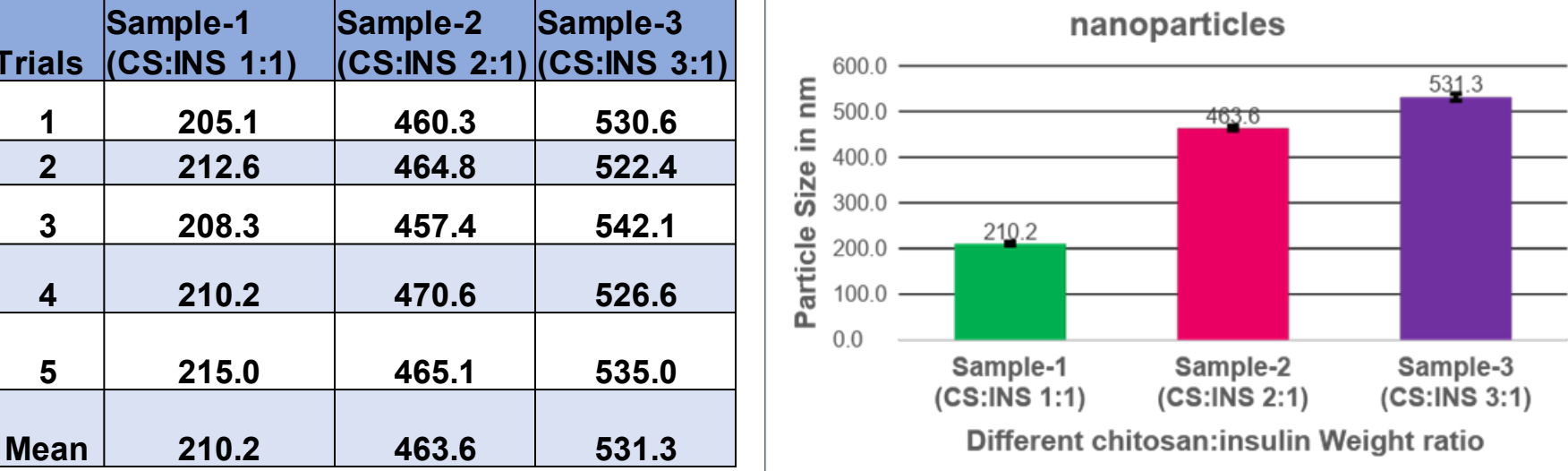


Table-2. Particle sizes of chitosan-insulin nanoparticles (In nm)

Zeta-Potential of Chitosan(CS)-Insulin(INS) nanoparticles (In mV)

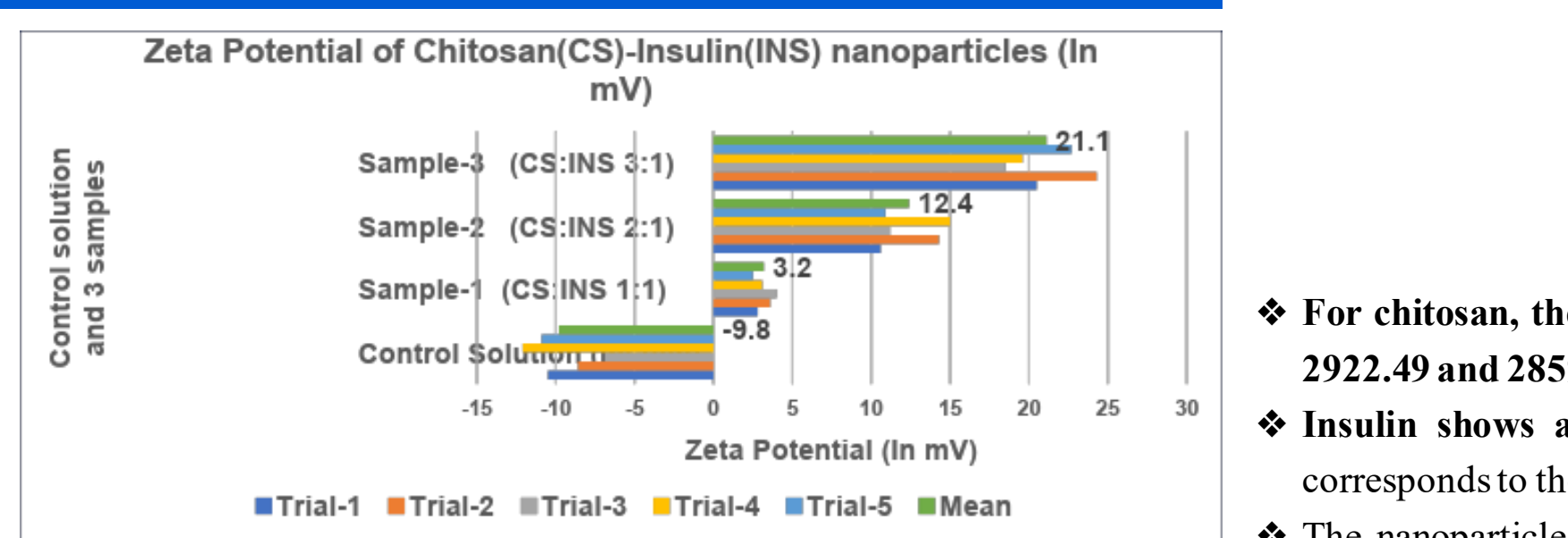


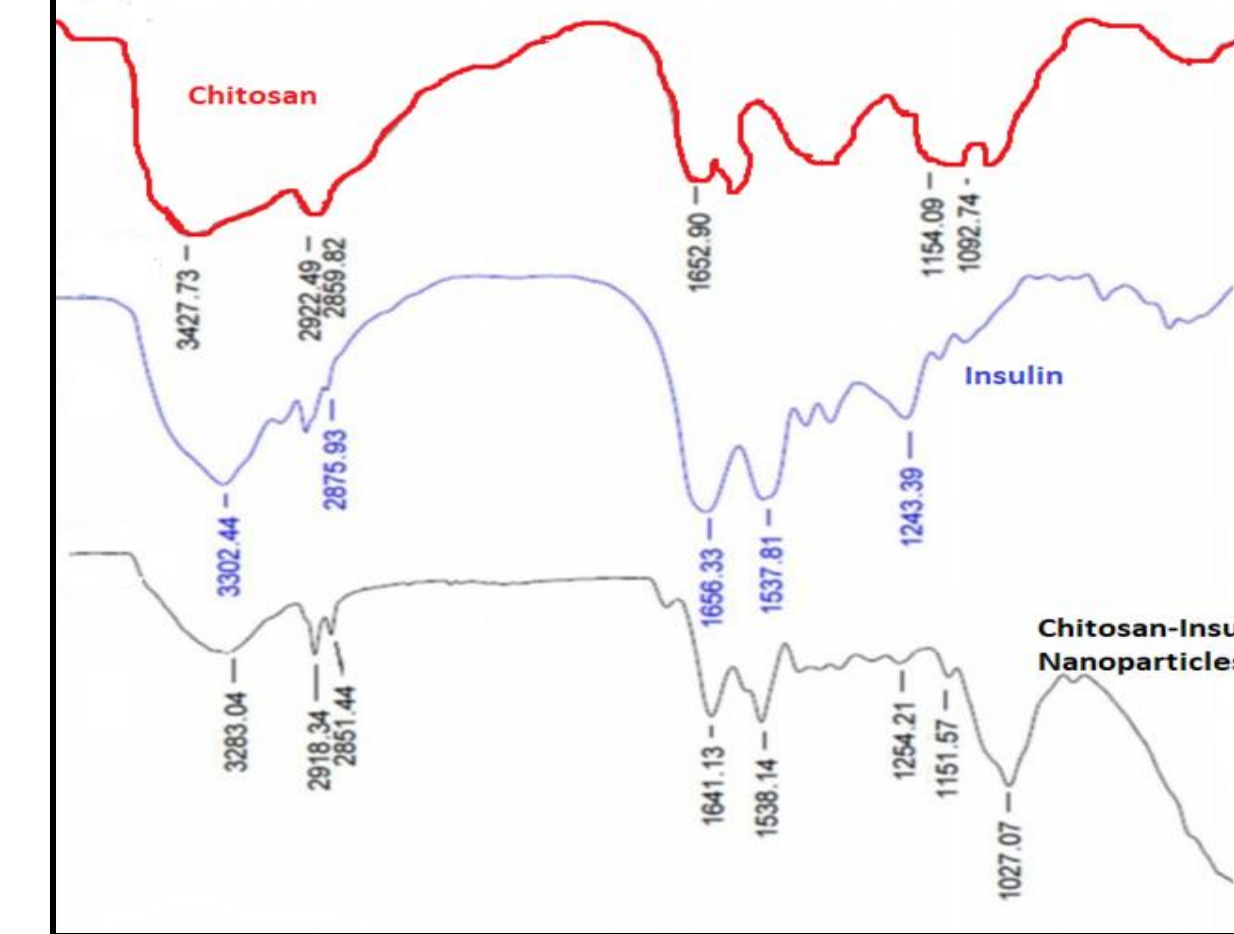
Table-3. Zeta Potential of Chitosan(CS)-Insulin(INS) nanoparticles (In mV)

Minutes	Stomach (pH 1.2)	Duodenum (pH 6.8)	Ileum (pH 7.4)
0	0	0	0
30	5.7	28.4	35.5
60	11.5	52.6	60.1
90	12.4	54.7	65.2
120	13.0	55.8	72.4
150	13.8	57.8	75.6
180	14.8	58.9	78.5
210	15.2	62.5	82.4
240	15.4	65.8	85.6

Table-5. Insulin release profile of the chitosan-insulin nanoparticles.

- The Insulin release profile were performed in the gastric (stomach, pH 1.2) and intestinal (duodenum, pH 6.8, ileum, pH 7.4) simulated pH conditions.
- The Insulin release is measured at 30 minutes intervals for 4 hours.
- at the acidic condition of stomach (pH 1.2), immediate release of insulin from the nanoparticles is only 15% of encapsulated insulin due to the strong ionic interaction.
- Almost entire (> 95%) amino groups of chitosan are protonated in lower pH protecting the entrapped insulin from enzymatic degradation in the stomach.
- In duodenum and ileum conditions, at pH 6.8 and 7.4, significant insulin is released from the nanoparticles. In pH 6.8 (duodenum), 65% insulin released in 4 hrs.
- Also, in the initial 1 hr., the insulin release was about 50%. This prolong release profile will help to produce desired effects of insulin release over long period.
- Similarly, at pH 7.4, mainly weak affinities between ionic groups of chitosan and insulin influenced prolonged and significant insulin release of almost 85%.

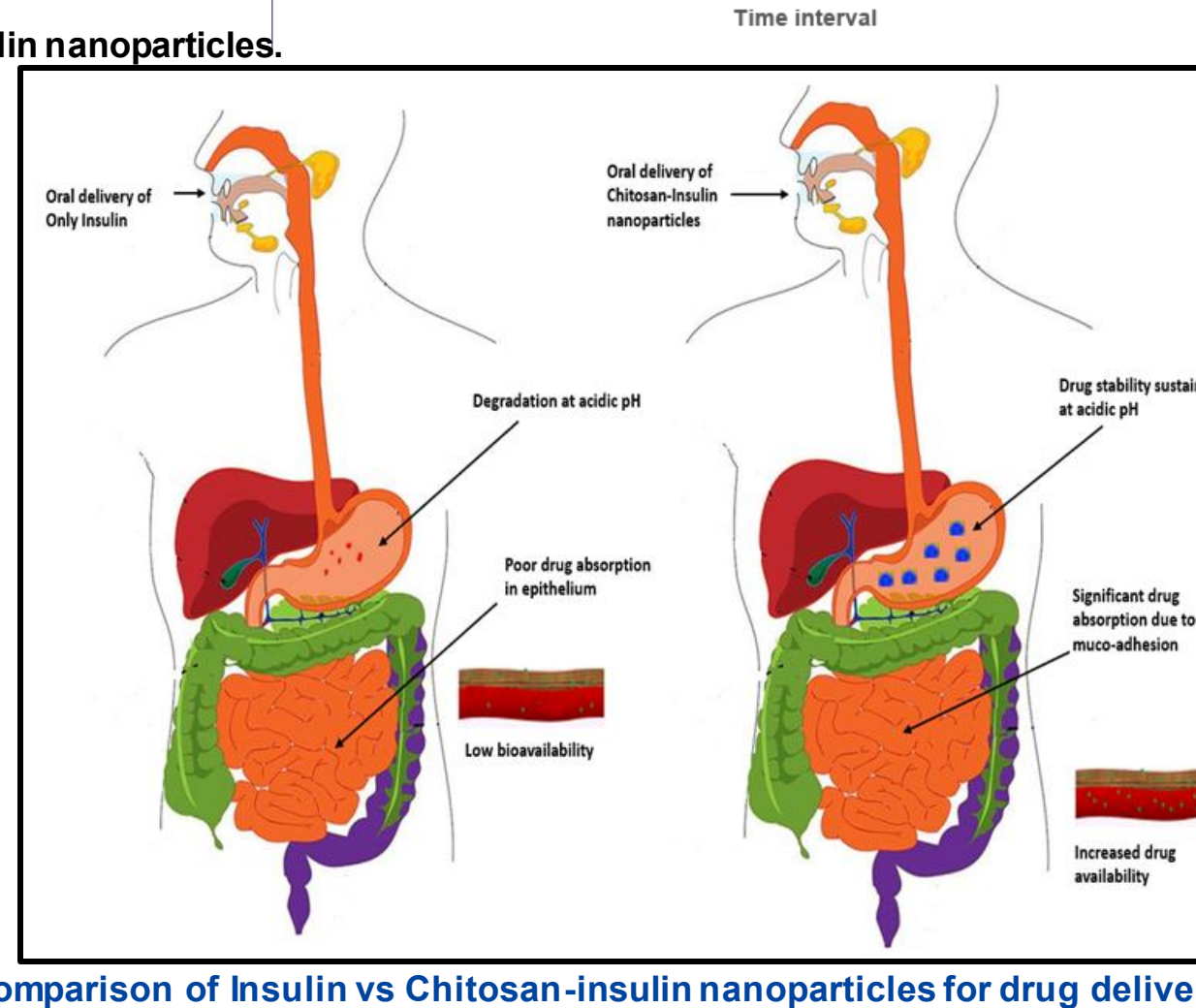
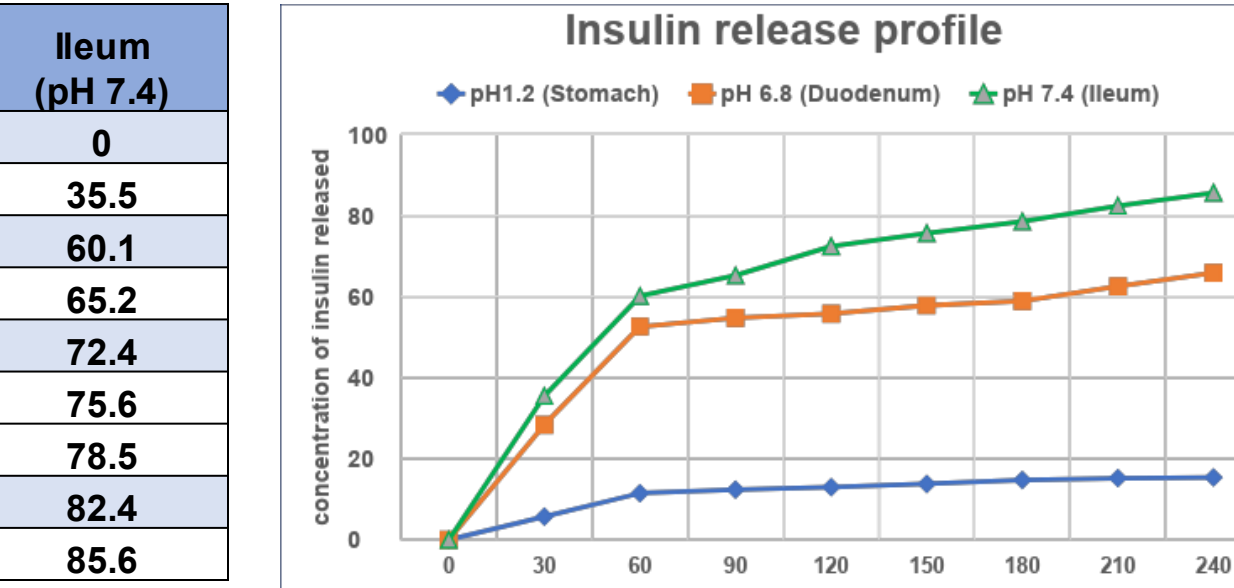
Fourier-Transform Infrared Spectrum



FT-IR Spectrum of Chitosan, Insulin and the chitosan-insulin nanoparticles

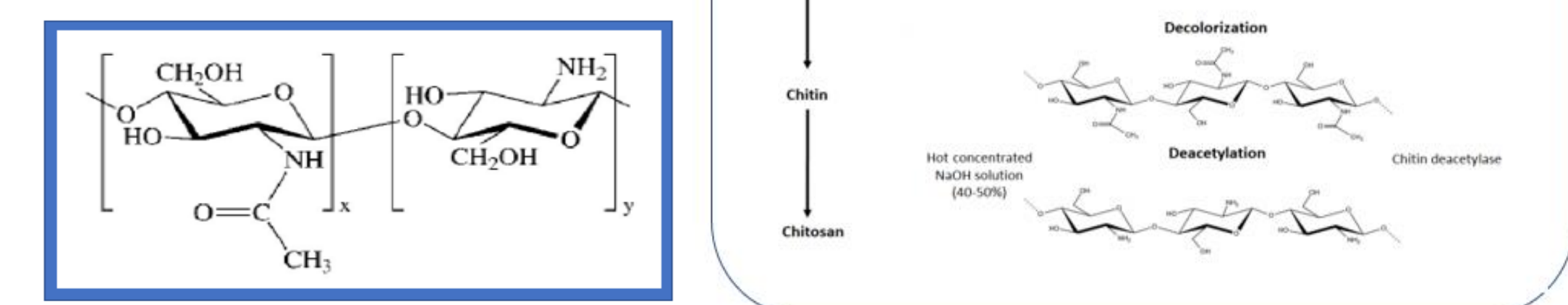
- For chitosan, the intensity peaks at 3427.73 cm⁻¹ (O-H stretch and N-H stretch, overlap), 2922.49 and 2859.82 cm⁻¹ (C-H stretch), 1652.90 cm⁻¹ (NH2 deformation)
- Insulin shows an intensity peak at 1656.33 cm⁻¹ (C=O stretching of amide I) which corresponds to the large α -helical of native insulin.
- The nanoparticles show little shift of amide I region towards lower spectra from 1656.33 to 1627.07 cm⁻¹.

In-Vitro Analysis of Insulin Release Profile



Chitosan and it's Properties

Chitosan is partially deacetylated form of chitin to at least 50% of the free amine form, has a heterogeneous chemical structure made up of both 1-4 linked 2-acetamido-2-deoxy- β -D-glucopyranose as well as 2-amino-2-deoxy- β -D-glucopyranose units on its structure.

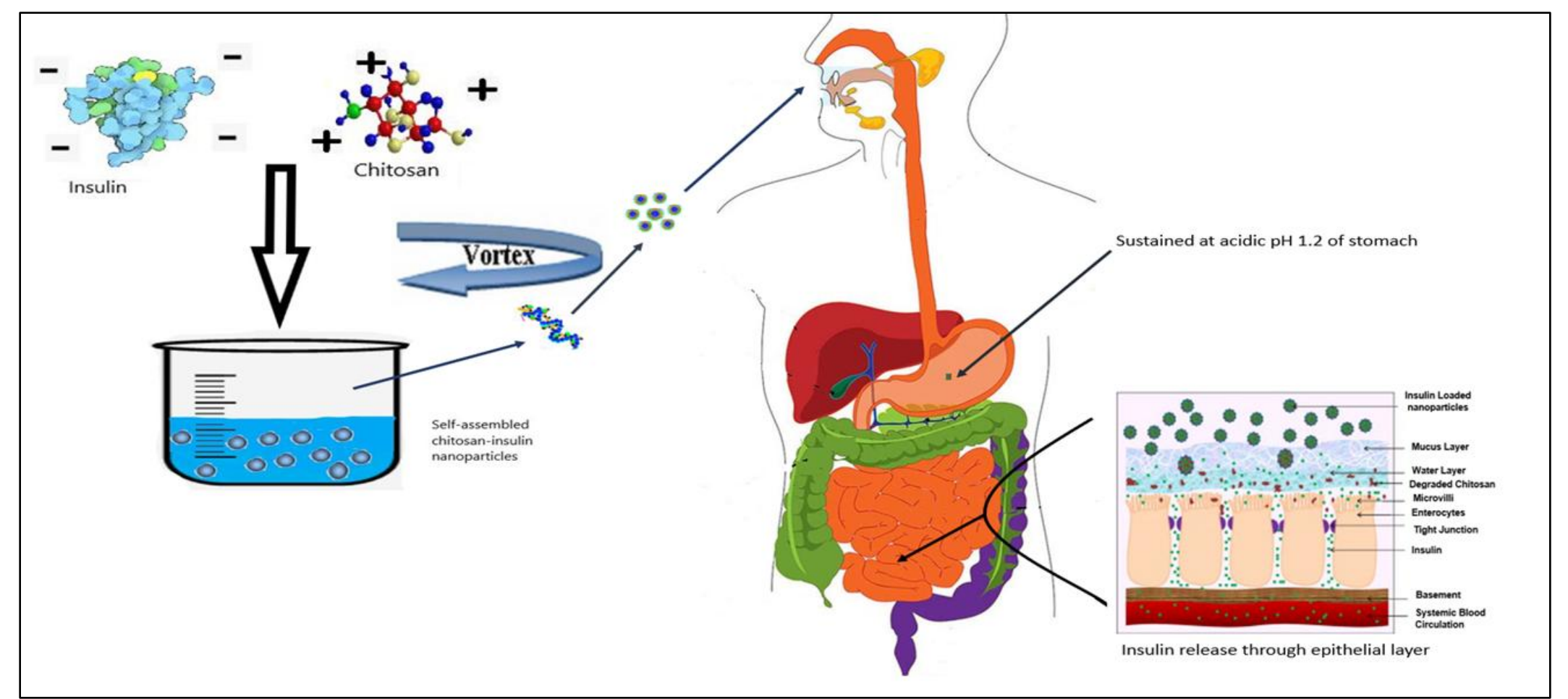


Chemical structure of Chitosan.

Chemical and Biological extraction of Chitosan.

Physicochemical properties of Chitosan

- It is a biodegradable, biocompatible, non-toxic and non-immunogenic muco-adhesive polymer. Thus, it offers advantages over other natural polymers in nanoparticle formation for oral drug administration.
- Chitosan is a cationic polymer due to protonation of amine groups in acidic pH possessing functional groups on its structure giving effective encapsulation of bio-molecules like proteins, drugs.
- Being a good muco-adhesive polymer, it can prolong the drug resident time in the GI tract.
- It also has excellent permeation enhancing and binding effects to epithelial for it's positive charges..



Use of the chitosan-insulin nanoparticles in oral insulin delivery

Toxicological Issues of Insulin Delivery Systems

- Toxicity is a critical factor to be considered when evaluating potential of chitosan-insulin nanoparticles.
- Nanoparticles are engineered to interact with cells, it is important to ensure that they do not cause any adverse effects or even damage the intestinal epithelium.
- The cytotoxicity test is used to detect the potential ability of nanoparticles to induce sublethal or lethal effects in cells. Chitosan was found as well tolerated, even at a dose 18 times higher than used in the study.

- The cytotoxicity was assessed in another study and it was proven that no extra toxicity was introduced.

Conclusion

- The objective of this project was in 3 parts:
 - Preparing the self-assembled chitosan-insulin nanoparticles for oral insulin delivery.
 - Determining its physicochemical properties of the prepared nanoparticles.
 - In-vitro analysis to determine the suitability of the nanoparticles for oral delivery of Insulin.
- The observations and data analysis clearly demonstrate that the hypothesis is fully accepted.
- The nanoparticles were prepared successfully by the self-assembling process without using any harsh organic chemicals that may be harmful for human body.
- The 1:1 ratio sample had average 210.2 nm particle size which is the smallest among 3 samples.
- The zeta potential show that chitosan has overbearing positive potential which can nullify the negative potential of insulin and still have positive charges which helps in muco-adhesion.
- All the samples obtained 79-93% insulin encapsulation efficiency.
- The in-vitro analysis of the insulin release profile demonstrates that the nanoparticle was able to achieve large insulin retention in the acidic pH condition (only 15% was released) of stomach.
- Significant amount insulin was released (up to 85%) in the intestinal pH condition.
- This in-vitro release profile show insulin retention in acidic pH, and most of the insulin was released in higher pH for sustained period of time which is suitable for oral insulin delivery.
- Based on the above results, this project concludes that the nanoparticles prepared by the self-assembling process demonstrated here without using harsh organic chemicals might be employed to improve the potential delivery approach for oral insulin, and chitosan-insulin self assembled nanoparticles can be potentially used for oral insulin delivery.
- Overall, this was a fun project to execute, and I learned about the different ways of drug delivery and the challenges that are associated with them. I also learned about the importance of nanotechnology and how it can be used to solve some major challenges.

Future Goal

- In addition to the current project, many other analyses would be necessary to determine the suitability in oral insulin delivery which I could not do due to constraints in the laboratory and approval processes:

- In-Vivo analysis using real animals, and study the effect of the nanoparticles on them.
- Toxicological analysis of the prepared nanoparticles, and determining of any harmful effects.
- Effect of external stimulants, temperature, pressure etc. and how those affects the nanoparticles.
- Determination of optimum dosage and its effect on reducing blood glucose levels over time.

- Therefore, the future goal of my project is to focus on these aspects of oral insulin delivery using the self-assembled nanoparticles.

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Introduction

Diabetes mellitus, a metabolic disease of digestive system, affects more than 100 million people around the world and appears mostly in two types: Type1 and Type2. It is a progressive decrease of β -cell function and when not or inappropriately treated, may lead to severe complications, which are life threatening and very costly to control. Therefore, the maintenance of blood glucose levels at near normal levels reduces the risk of long-term complications of diabetes, such as adult blindness, cardiovascular diseases, non-traumatic amputation, and diabetic nephropathy. In fact, intensive insulin in Type-1 patients is able to reduce the risks of nephropathy by 35% to 56%, neuropathy by 60%, and retinopathy by 50% to 70%.

Insulin is usually administered by the subcutaneous route, which significantly reduces morbidity and mortality. However, approximately 60% of patients fail to achieve long-term glycemic control. This may be due to poor patient compliance owing to the use of needles and the complexity of the insulin treatment regimen, the late stage at which insulin may be prescribed, and fear of hypoglycemia episodes and weight gain. To overcome such problems, different routes of insulin administration are being tested. The oral route remains the preferred choice for drug administration because of its noninvasive nature. However, proteins such as insulin have low oral bioavailability due to their intrinsic lack of permeability through the intestinal epithelium. Therefore, to develop a delivery system intended to provide insulin orally, a proper understanding about the mucosal micro-environment and intestinal physiology is required.

During past years, different drug delivery strategies have been introduced to overcome the low oral bioavailability of insulin. Recently, nanoparticles drew a lot of attention of the pharmaceutical scientists in drug delivery research due to its versatility in targeting tissues, enabling deep molecular targets as well as sustained drug release. Apart from drug delivery, nanoparticles displayed excellent efficiency as potent proteins, peptide, vaccines and DNA carrier. Thus, nanoparticles offer a better way for delivering these hydrophilic drugs as well as macromolecules following different routes of administration. Among several routes (nasal, oral, pulmonary rectal, and ocular) of drug administration, oral is the most preferred one, producing maximum patient response. But, still oral

deliveries of many therapeutic agents remain unsuccessful due to large particle size, hydrophilicity and instability within the animal body. During passage through oral route, drug molecules encountered with several barriers, like rapid enzymatic degradation and poor intestinal absorption, limiting their potential to reach the blood stream for exerting therapeutic functions.

Chitosan is a natural polymer derived from chitin by partial deacetylation. Chitin is a natural bio-polymer generally found in crustacean (crabs, shrimps and lobsters) shell and in some fungi or yeast. Chitosan is linear polymer having (1-4)-linked 2-amino-2-deoxy- β -D-glucopyranose units on its structure. It is biodegradable, biocompatible, non-toxic and non-immunogenic muco-adhesive polymer. So, it offers certain advantages over other natural polymers in formulating nanoparticles for oral administration of drug molecules. Chitosan is a cationic polymer (due to protonation of amine groups in acidic pH) possessing functional groups on its structure facilitating effective encapsulation of bio-molecules like proteins, drugs. Being a good muco-adhesive polymer, chitosan can prolong the drug resident time in the gastrointestinal tract. The biocompatibility and biodegradability of chitosan render it as suitable candidate for the insulin delivery carrier.

In the current project it is aimed to make the self-assembly of Chitosan and Insulin nanoparticles, and use that for oral insulin delivery. Therefore, it aims to encapsulate insulin into chitosan by self-assembly method and to observe the morphological characters of the synthesized nanoparticles. Also, it aims to determine the effectivity of Chitosan nanoparticles as the carrier of Insulin and how does self-assembled Chitosan-Insulin nanoparticles improve the insulin retention and release at different pH levels in the gastrointestinal (GI) tract.

Objectives

There are 3 main objectives of this project:

- 1. Make a self-assembly of Chitosan and Insulin nanoparticles by encapsulating insulin into chitosan without using any harsh chemicals.**
- 2. Determination of the physicochemical properties of the Chitosan-Insulin nanoparticles.**
- 3. In-Vitro analysis of the nanoparticles for Insulin encapsulation efficiency and Insulin release profile in Human digestive system pH conditions.**

Question, Hypothesis and Variables:

Question:

How do different weight ratios of Chitosan-Insulin nanoparticles improve insulin retention and release at different pH corresponding to the human digestive system GI tract?

Hypothesis/Engineering Goals:

Hypothesis: If 3 samples of different weight ratios of Chitosan and Insulin are taken in the self-assembled Chitosan-Insulin nanoparticle,

Sample No.	Chitosan:Insulin weight Ratio
1	1:1
2	2:1
3	3:1

their particle sizes are analyzed and they are put in the pH conditions corresponding to the human digestive system, then the Chitosan-Insulin weight ratio 1:1 will be able to achieve the lowest particle size and all the weight ratios of the self-assembled nanoparticles will be able to achieve more concentration of released Insulin than the control sample of only insulin without any Chitosan nanoparticle carrier.

Variables :

Independent variable (IV):

3 different weight ratios of Chitosan and Insulin nanoparticles.

Dependent variable (DV):

The Concentration of the released Insulin.

Constants:

The pH conditions, Volume of the samples, Degree of deacetylation of the samples, Temperature, etc.

Background Research

Type-1 and Type-2 Diabetes

Diabetes Mellitus, or more commonly known as diabetes, is a chronic condition that occurs when there are signs of abnormally high blood glucose or blood sugar levels, also known as hyperglycemia. This usually takes place when the pancreas produces little to no insulin or when the body cells cannot use the insulin properly. Insulin is a protein hormone which regulates blood sugar levels, it allows the body to use glucose from carbohydrates in the food eaten for energy, or to store it for future use. Insulin keeps blood sugar levels from getting too high or low. There are two main types of diabetes. Type 1 diabetes takes place when the pancreas doesn't produce any insulin at all due to damage in the cells where insulin production takes place. Approximately 5% - 10% of the population have type 1 diabetes. Type 2 diabetes occurs due to a low production of insulin or the body does not effectively use the produced insulin. This is much more common within diabetes sufferers, with about 90% - 95% having type 2 diabetes. There are other types of diabetes as well such as gestational diabetes that occurs from pregnancies and diabetes from diseases in the pancreas. However, they are not as common as Type 1 and type 2 diabetes.

Current treatment with Insulin

Insulin, a protein hormone secreted by pancreatic β -cells, plays a crucial role in the regulation of glucose metabolism and is used as a drug to control Type-1 and Type-2 Diabetes Mellitus. Insulin is usually administered by the subcutaneous route as injection, which significantly reduces morbidity and mortality. However, approximately 60% of patients fail to achieve long-term glycemic control. This may be due to poor patient compliance owing to the use of needles and the complexity of the insulin treatment, the late stage at which insulin may be prescribed, and fear of hypoglycemia episodes and weight gain. To overcome such problems, different routes of insulin administration are considered. The oral route remains the preferred choice for drug administration because of its noninvasive nature. However, proteins such as insulin have low oral bioavailability due to their intrinsic lack of permeability through the intestinal epithelium. Therefore, to develop a delivery system intended to provide insulin orally, a proper understanding about the mucosal microenvironment and intestinal physiology is required.

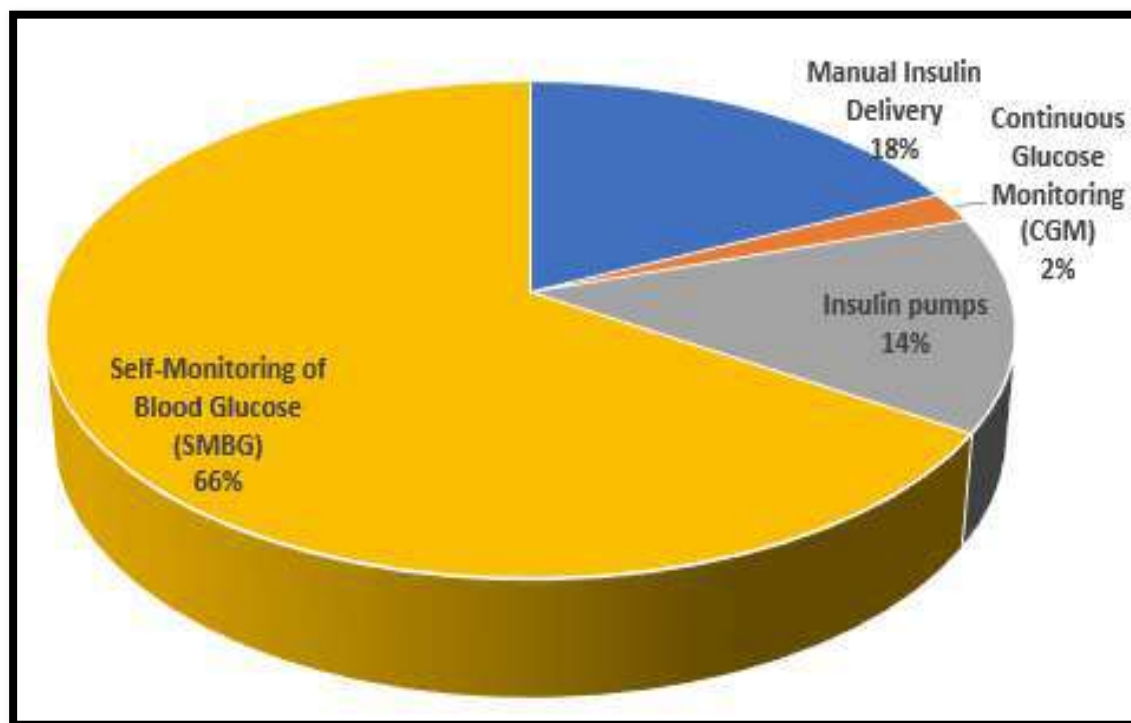


Figure- 1: Current Market share of Insulin devices.

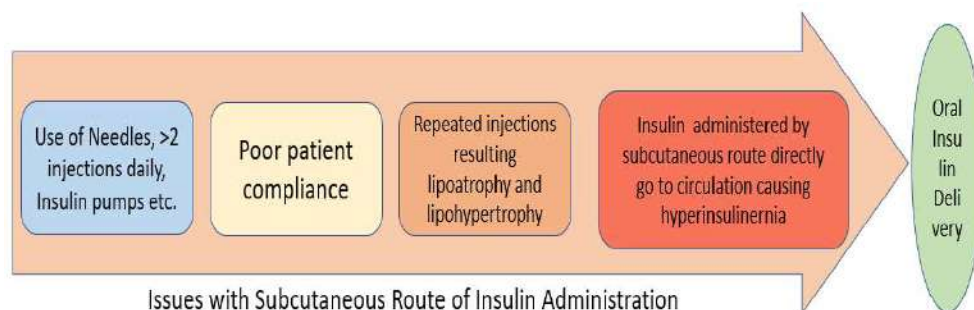


Figure- 2: Issues with Subcutaneous route of Insulin administration.

Advantages and Disadvantages of Oral Insulin Delivery

Insulin is a protein or peptide hormone produced by the β -cells of the pancreatic islets. High molecular weight and hydrophilicity of proteins hinder their intestinal absorption, leading to low oral bioavailability, negligible plasma levels, and high variability. Thus, therapeutic proteins are commonly administered by parenteral route, therefore, insulin being a peptide, is administered subcutaneously in the treatment of diabetes mellitus. However, oral administration is regarded as a better route of administration because of its cost-effectiveness and well-established acceptability, and especially because it allows avoiding the use of needles and other injection materials. It also reduces contamination risks, local pain, and immune reactions and have better patient compliance. Also, oral administration of insulin better mimics the normal insulin pathway in the body after endogenous secretion, providing a better glucose homeostasis. In fact, oral delivery of insulin allows high portal vein concentrations, with no sustained peripheral hyperinsulinemia, which is associated with neuropathy and retinopathy.

However, oral delivery of insulin has some serious limitations, which include low bioavailability due to insulin degradation in the gastrointestinal tract (GI Tract) by proteolytic enzymes and severe pH physiological conditions as well as poor permeability through the intestinal epithelium. During passage through oral route, drug molecules encounter with several barriers, like rapid enzymatic degradation and poor intestinal absorption, limiting their potential to reach the blood stream for exerting therapeutic functions.

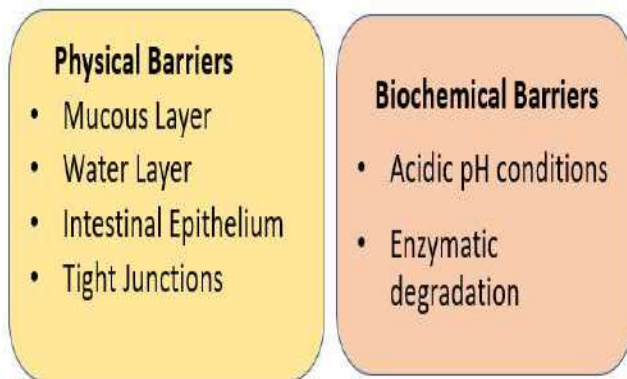


Figure- 3: Barriers in Oral Insulin Delivery.

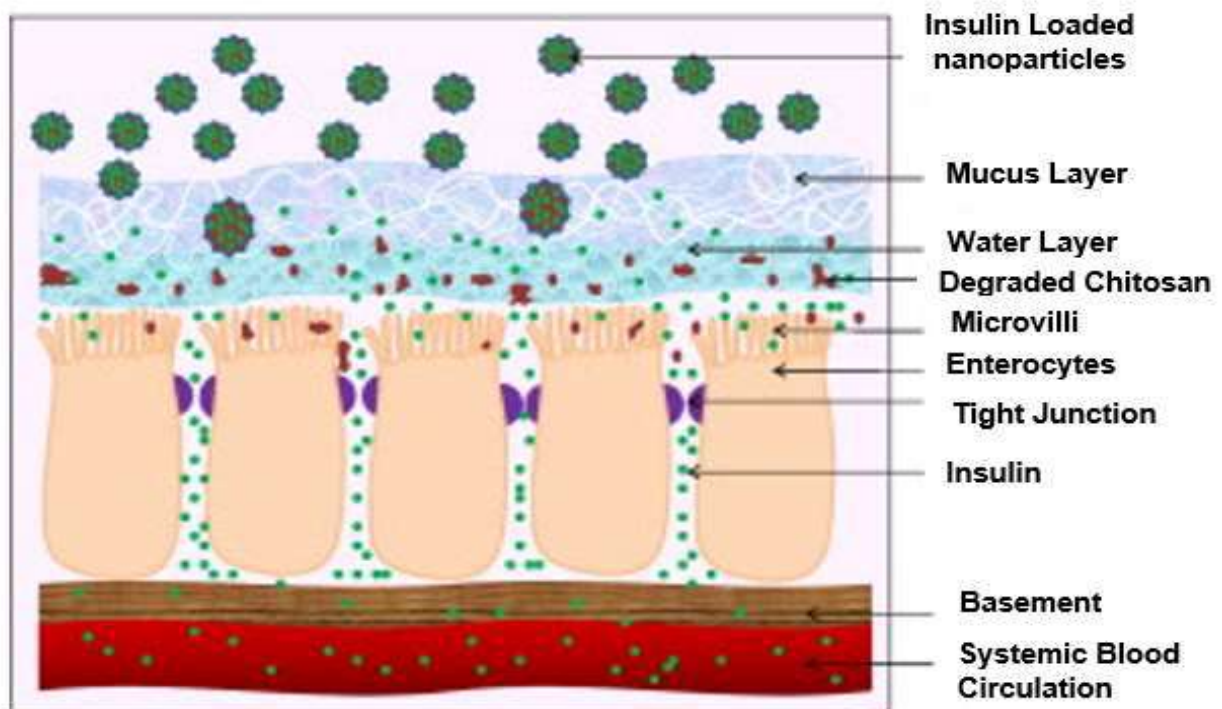


Figure- 4: Insulin release through the epithelial layers.

Delivery Systems for Oral Insulin Administration

To develop an adequate oral delivery system, the sustenance of insulin's biological stability in the GI Tract must be addressed during the formulation process. Several insulin delivery systems, such as tablets, capsules, intestinal patches, hydrogels, microparticles,

and nanoparticles, have been explored to deliver insulin by paracellular and/or transcellular transport throughout the ileum and colon. The delivery systems used should contain excipients, which protect insulin from aggregation and enzymatic degradation, prolong its residence time in the GI Tract, and enhance its intestinal uptake.

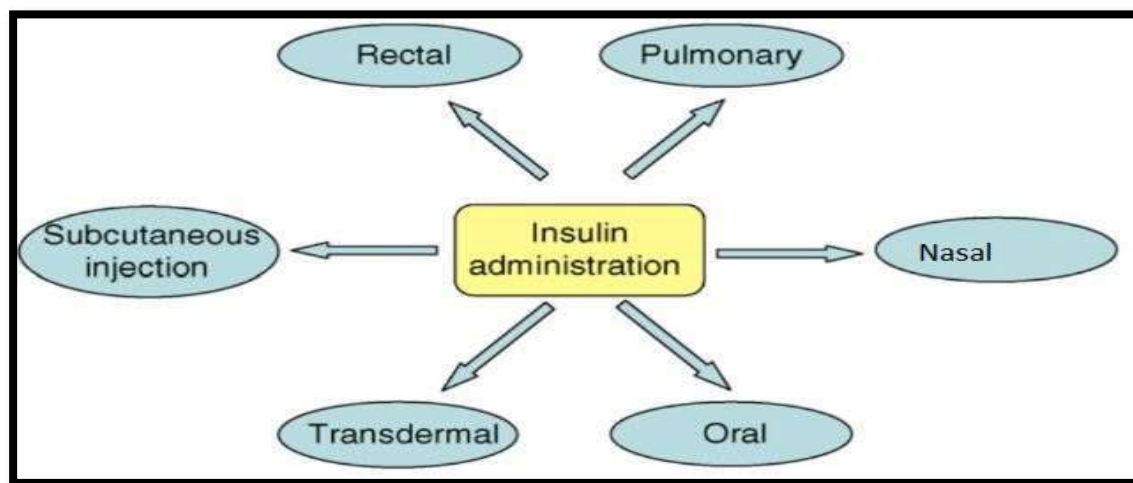


Figure- 5: Different routes of Insulin administration.

Use of Nanotechnology in oral delivery of Insulin

Recent advances in biotechnology and genetic engineering have made the mass production of therapeutic micro-molecules, such as proteins, nucleic acids and polysaccharides, feasible. The nanoparticles draw a lot of attention of the pharmaceutical scientists in drug delivery research due to its versatility in targeting tissues, enabling deep molecular targets as well as sustained drug release. Apart from drug delivery, nanoparticles display excellent efficiency as potent proteins, peptide, vaccines and DNA carrier. Thus, nanoparticles offer a better way for delivering these hydrophilic drugs as well as macromolecules following different routes of administration. Several natural polymers like, alginate, pectin, gelatin, chitosan, etc. are commonly used to prepare the nanoparticles.

What is Chitosan

Chitosan is a natural polymer derived from chitin by partial deacetylation. Chitin is a natural bio-polymer generally found in crustacean (crabs, shrimps and lobsters) shell and in some fungi or yeast.

Chitosan, deacetylated form of chitin to at least 50% of the free amine form, has a heterogeneous chemical structure made up of both 1-4 linked 2-acetamido-2-deoxy- β -D-glucopyranose as well as 2-amino-2-deoxy- β -D-glucopyranose units on its structure.

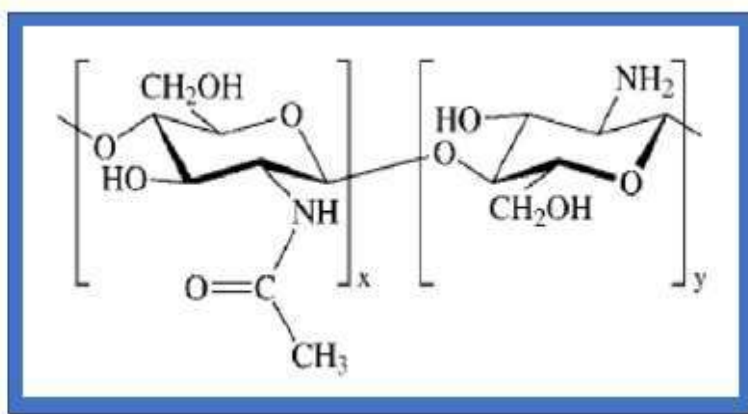


Figure- 6: Chemical structure of Chitosan.

It is biodegradable, biocompatible, non-toxic and non-immunogenic mucoadhesive polymer. So, it offers certain advantages over other natural polymers in formulating nanoparticles for oral administration of drug molecules. Chitosan is a cationic polymer due to protonation of amine groups in acidic pH possessing functional groups on its structure facilitating effective encapsulation of bio-molecules like proteins, drugs. Being a good muco-adhesive polymer, chitosan can prolong the drug resident time in the GI tract. Chitosan has been shown to have excellent permeation enhancing effects and binding effects to epithelial cells due to the positive charges on its structures.

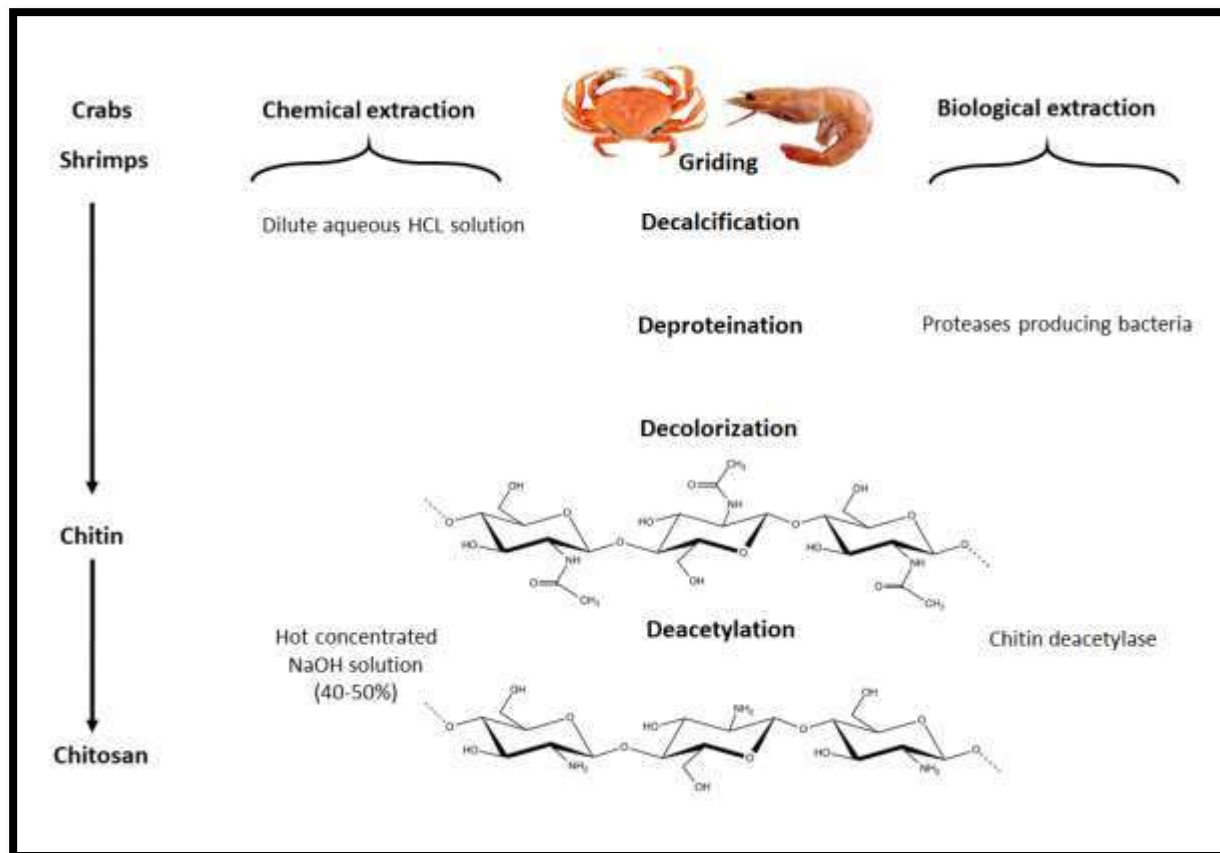


Figure- 7: Chemical and Biological extraction of Chitin and Chitosan.

Different Techniques for Chitosan Nanoparticle preparation

There are several techniques for synthesizing chitosan nanoparticles such as ionotropic gelation, micro emulsion method, Emulsification solvent diffusion, solvent evaporation, Coprecipitation and complex coacervation method. Most of these methods except complex coacervation method involving organic solvents, vigorous agitation process with heat generation limit their use in synthesizing chitosan nanoparticles. These chemicals and harsh procedures can be potentially harmful to the sensitive bio-molecules like protein, peptide drugs, and DNA etc.

Self-Assembled Chitosan-Insulin Nanoparticles

The current project proposes preparation of self-assembled Chitosan-Insulin nanoparticles. Self-assembly of proteins or peptide drugs with natural or synthetic polymers are possible. This process leads to the formation of polyelectrolyte-complexes by electrostatic interaction, which is optically homogeneous, stable nano-dispersion at colloidal level. Mainly, poly-electrolytes are ionizable group containing polymers, which can form poly-electrolytic complex with opposite charged bio-molecules via intermolecular interactions. The interactions may be either Vander Waals force, hydrogen bonding, coulombic forces or transfer forces. This self-assembly technique of nanoparticle synthesis can be advantageous over others by lowering the risk of the damage of entrapped drug molecule as it involves no organic reagent or any harsh method of nanoparticles preparation. This project proposes the use chitosan nanoparticles for oral insulin delivery. Therefore, it aims to encapsulate insulin into chitosan by self-assembly method and to observe the morphological characters of synthesized nanoparticles. In addition, it aims to investigate the in-vitro release of insulin in different simulated pH conditions of human digestive system.

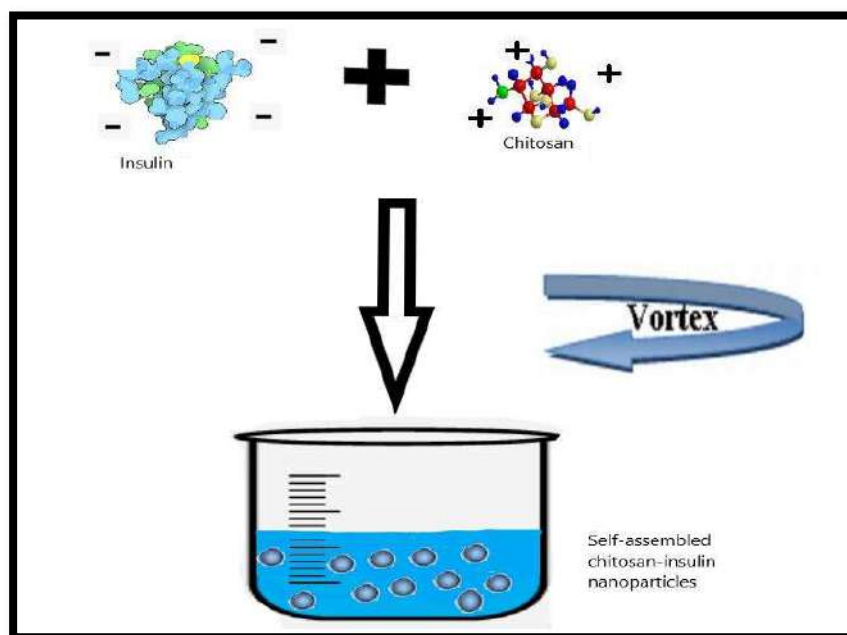


Figure-8: Self-assembly of Chitosan-Insulin Nanoparticles.

Main factors affecting physicochemical properties of Chitosan

The two most important factors that determine the physio-chemical properties, and consequently the specific applications of chitosan, are chitosan's degree of deacetylation (DD) and molecular weight (MW). DD is the percentage of the deacetylated units (glucosamine monomers) in chitosan's chains, it affects the chemical, physical and biological properties of chitosan, such as the tensile strength of the films, the ability to chelate metal ions and the immuno-adjuvant activity. DD also affects the intrinsic pKa of chitosan, leading to the alteration of chitosan's solubility in dilute acidic solutions. Generally, $DD \geq 40\%$ are soluble in dilute acidic solutions. Protonation of the amino groups of glucosamine units contribute to the disruption of hydrogen bonding, the solvation of cationic sites and then to the solubilization of chitosan when the balance between solvent/polymer and polymer/polymer interactions becomes favorable. When the DD becomes less than 40%, chitosan chains become completely insoluble in water, due to

- 1) the numerous H-bonds that occur between alcohol, amide and ether functionalities distributed on the repeating units all along the polymer chains and
- 2) the hydrophobic interactions due to the presence of the methyl groups of the acetamide functions and to the $-CH$ and $-CH_2$ of the glucosidic rings.

Toxicological Issues of Insulin Delivery Systems

Toxicological issues must be focused on the binomial insulin carrier. Toxicity is a critical factor to be considered when evaluating the potential of insulin-loaded nanoparticles. Given that nanoparticles are engineered to interact with cells, it is important to ensure that they do not cause any adverse effects or even damage the intestinal epithelium. The important issue is that, whether naked or coated, nanoparticles will undergo biodegradation in the cellular environment and may affect cellular responses. For instance, biodegraded nanoparticles can accumulate inside the cells and lead to intracellular changes, such as disruption of organelle integrity or gene alterations, which cause severe toxicity. The cytotoxicity test is a sensitive, rapid, and inexpensive method to detect the potential ability of nanoparticles to induce sublethal or lethal effects in cells.

However, cytotoxicity may not be the only adverse effect because cell immunological response may also be affected. Furthermore, molecules delivered to unnatural sites in unnatural quantities are likely to behave in unexpected ways, so from a toxicological perspective, oral insulin delivery may be questionable. If insulin is entrapped and not released from carrier systems until it reaches the systemic circulation, then this may not be an issue, but this approach is questionable, because insulin may cause gastroparesis. The use of absorption enhancers may lead to a long-term toxicity, and surfactants can also damage intestinal epithelium. Indeed, absorption enhancers, when administered in a continuous manner, may also promote permeation of pathogens and toxins.

In another study, the safety of self-assembled nanoparticles, obtained mixing a poly- γ -glutamic acid solution with chitosan in presence of $MgSO_4$ and sodium TPP, after oral administration in a diabetic rat model were assessed by an *in vivo* toxicity study. These nanoparticles were well tolerated, even at a dose 18 times higher than used in the pharmacodynamic/pharmacokinetic study. Another biocompatibility assessment of N-trimethyl chitosan–cysteine nanoparticles when administered to rats revealed lack of toxicity.¹⁰² Moreover, the cytotoxicity of PLGA nanoparticles was assessed by 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide assay, and it was proven that no extra toxicity was introduced by nanoparticles.

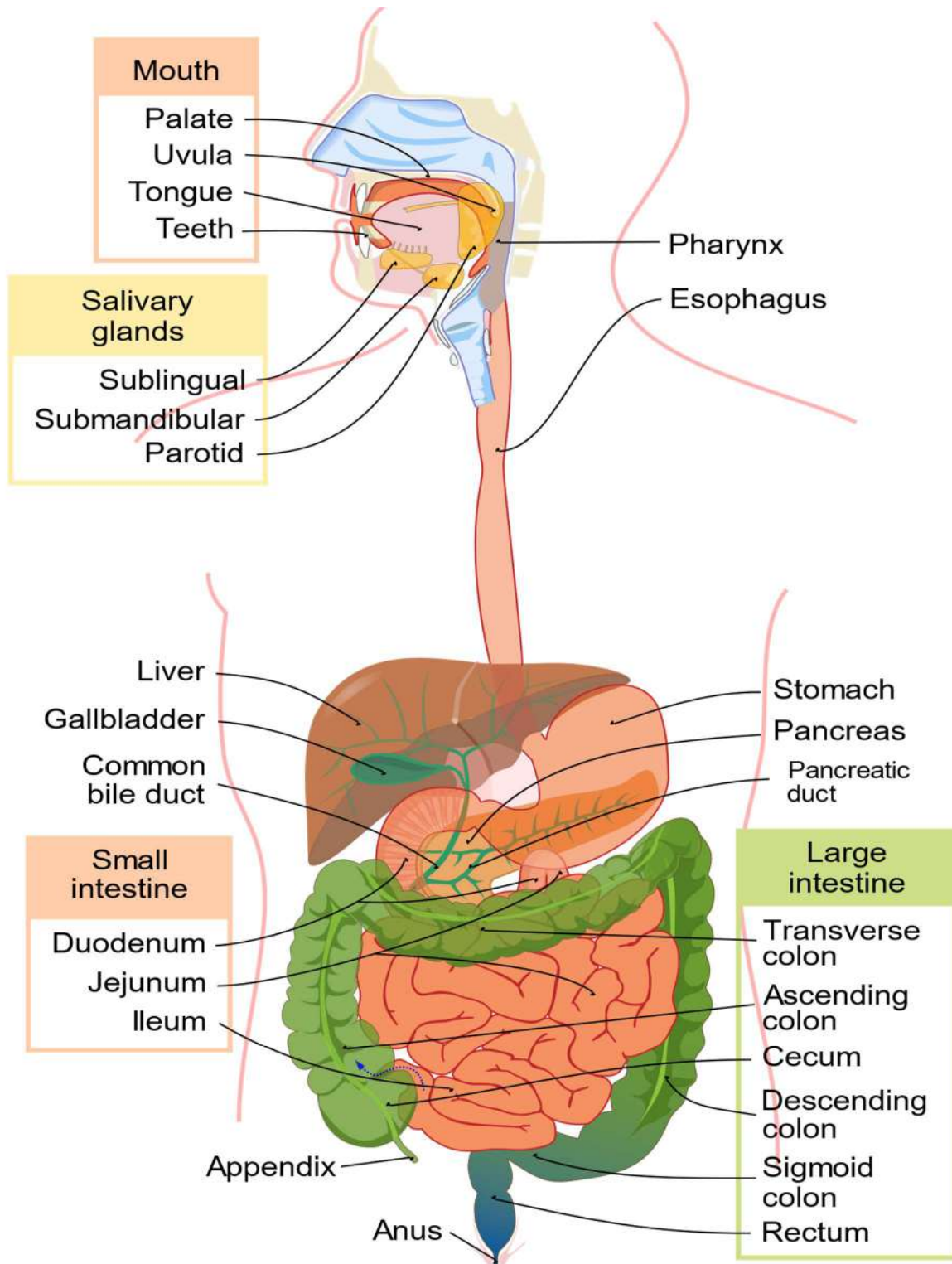


Figure-9: Parts of Human digestive system.

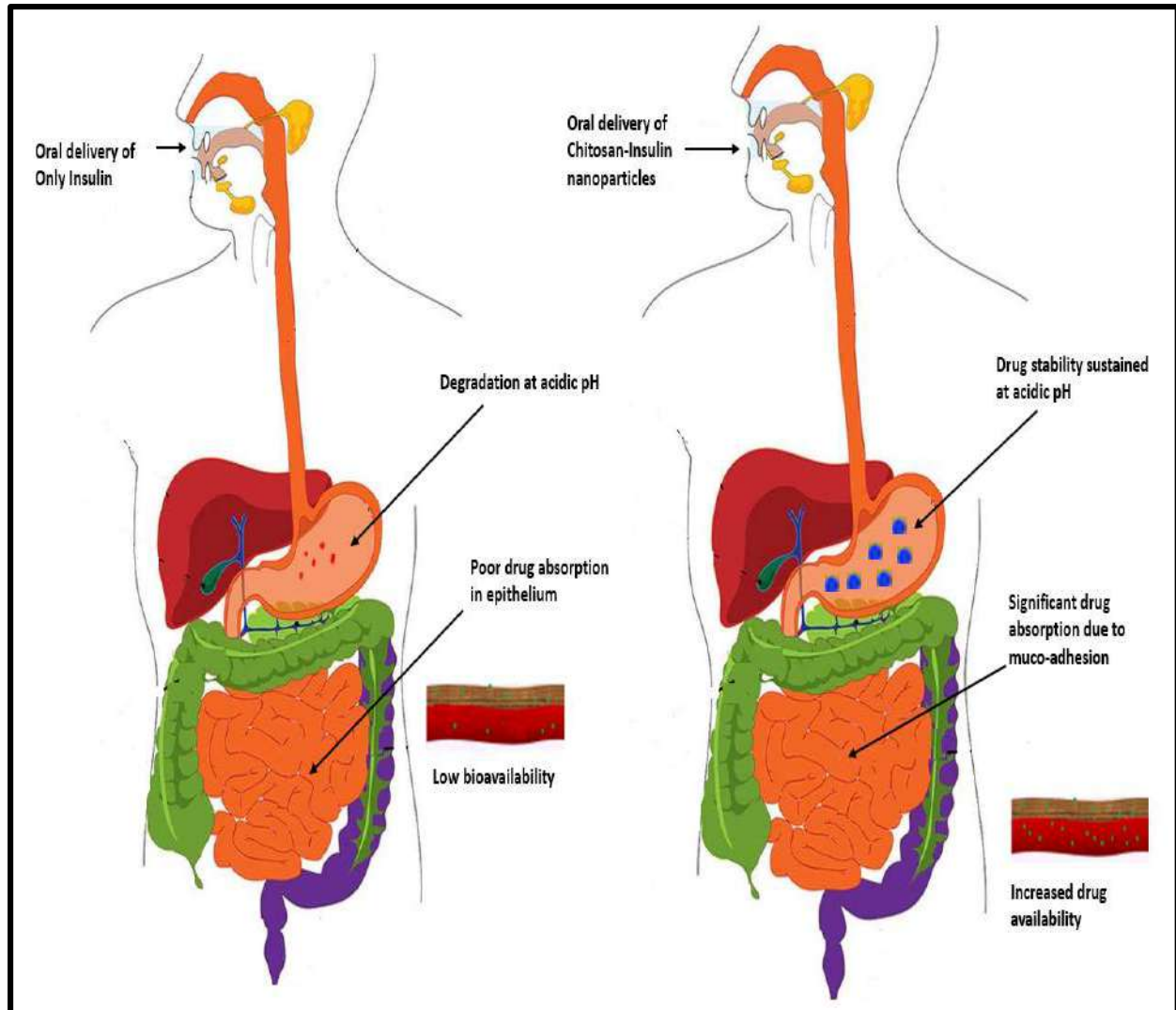


Figure-10: Comparison of oral delivery of Insulin and the chitosan-insulin nanoparticles through digestive system.

Materials and Methods

Materials:

Glacial Acetic Acid. 60.052 g/mol 1.049 g/cm³

Tris (hydroxymethyl) Aminomethane 500 g

Chitosan 70-150 mPa.s

Insulin 27 USP units/mg

Distilled Water 10 L

Erlenmeyer flasks 500 ml - 8 Nos.

Graduated cylinder 100 ml – 4 Nos.

Glass funnels and filter stand with Burette clamp.

Test tubes - 20 nos.

Hydrochloric Acid - 30 ml

Milipore 0.5 µm filter

Sodium Hydroxide (NaOH) solution -100 ml

Sodium Nitrite (NaNO₂) - 50 gm

Portable pH meter.

Low Profile Vortex Mixer.

UV-vis Spectrophotometer.

Infrared Spectroscope.

Zetasizer Nano ZS.

Glass Capillary Viscometer.

Experiment Procedures

1. Preparation of the Nanoparticles.

The biocompatibility and biodegradability of chitosan render it as suitable candidate material for medical applications. Two main aspects,

- i) Molecular Weight (MW) and
- ii) Degree of deacetylation (DD)

of chitosan plays a significant role for determining the suitability of chitosan for oral drug delivery. These affects the chemical, physical and biological properties of chitosan, such as DNA condensation and release, immune-adjuvant activity and cellular uptake.

To prepare the Chitosan based Nanoparticles suitable for oral drug delivery, the following steps were taken.

2. Purification of Chitosan

The low molecular weight chitosan will be prepared by oxidative degradation with Sodium Nitrate NaNO_2 at room temperature using the following procedure.

- a) Make 1% acetic acid solution by adding 2 ml 99.8% Glacial acetic acid in 198 ml distilled water.
- b) Weigh 2 gm of dry chitosan and pour 200 ml of the 1% acetic acid solution in it.
- c) Stir the solution to dissolve the chitosan to make 1%(w/v) chitosan dissolved in 1% acetic acid.
- d) Add 10 ml of NaNO_2 in 1% acetic acid drop wise for 30 min and stir it vigorously.
- e) Keep the mixture at room temperature for 3 h for the reaction to complete.
- f) Neutralize the reaction mixture with 1N NaOH to pH 7.4 to complete precipitation of chitosan.
- g) Wash the precipitate with distilled water, and dry it overnight at 55°C to obtain purified chitosan.

3. Determination of the Molecular weight (MW) of Chitosan

The molecular weight of chitosan is determined by Glass Capillary Kinematic Viscometer.

- Stabilize temperature of the water bath at 25°C and monitor for 10 min before the first measurement.
- Filter the 1% chitosan solution, twice through a 0.5 µm filter.
- Place 10 ml of the filtered chitosan solution in the viscometer. Measure the flow time for 5 times.
- Add 10 ml of the filtered solvent was to the viscometer and measure the flow time of the resulting diluted solution 5 times.
- Dilute the solution by adding 10 ml solvent each time for 5 times. Due to the pressure difference at the top and bottom of the column of liquid, the capillary corrections of flow times according to capillary type is introduced in the calculations, using the formula provided by the capillary manufacturer.

4. Determination of degree of deacetylation (DD) of Chitosan

The degree of deacetylation (DD) of chitosan is determined by titration method.

- Dissolve 1 g of chitosan in 250 ml of 0.1 mol/L HCl solution.
- Add 100 ml distilled water and calculated amount of NaCl to adjust the ionic strength to 0.1 mol/L.
- Perform titration by adding the titrant solution of 0.1 mol/L NaOH to the chitosan solution gradually.
- Record the volume of NaOH added and pH values using digital pH-meter.

DD (%) of chitosan will be calculated by using the following equation:

$$DD (\%) = \frac{203.2}{42.0 + \frac{1000m}{C_{NaOH}(V2 - V1)}} \times 100$$

where m is the amount of chitosan (gm) in the solution, NaOH c is the concentration of NaOH solution (M), V2 – V1 is the volume of NaOH solution consumed in titration of

amino groups of chitosan, 203.2 is the molecular weight of the acetylated monomeric unit of the polysaccharide, 42.0 is the difference between the molecular weights of the deacetylated monomeric unit and acetylated monomeric unit, 1000 is the conversion factor of milliliters to liters and 100 is the conversion factor of DD to percent.

5. Preparation of self-assembled Chitosan-Insulin Nanoparticles

For the formation of the self-assembled Chitosan-Insulin nanoparticle, first we need to obtain purified Chitosan from Step 2.

- a) Dissolve the dry purified chitosan made before in 1 % acetic acid solution and stir for 30 min.
- b) Adjust the pH of the solution at 5.5 -5.7 by the addition of sodium hydroxide solution.
- c) Dissolve lab-grade Insulin (27 USP/ml) in 0.1M HCl solution.
- d) Adjust the pH of the Insulin solution to 8.4 by 0.1 N Tris (hydroxymethyl) aminomethane solution.
- e) Mix equal volume of both chitosan and insulin solutions and with a Vortex mixer, vortex for 30 sec.
- f) Incubate the mixture at room temperature for 30 min to completely form the self-assembled complexes.

6. Analyze the Characteristics of the self-assembled Nanoparticles:

To analyze the physical characteristics of the self-assembled Nanoparticles obtained by the abovementioned method, the following analysis will be done:

6.1 Fourier transform infrared (FT-IR) spectroscopy analysis

- a) Fourier transform infrared (FTIR) analysis is done using ATR-FTIR spectrometer.
- b) Scan from 4000 to 500 cm^{-1} at room temperature.
- c) For the FT-IR spectroscopy analysis, use the Chitosan solution, insulin solution and chitosan/insulin self-assembled nanoparticles that were mixed separately.
- d) Record the measurements.

6.2 Determination of particle size and zeta potential

- a) The particle sizes and distributions of the self-assembled chitosan-insulin nanoparticles were determined by Dynamic light Scattering (DLS) scan.
- b) To avoid the influence of dust on the reliability of results, the solutions of the chitosan/insulin complex were filtered through a 0.5 µm filter (Millipore) directly into a freshly cleaned 10 mm diameter cylindrical cell.
- c) Measure the intensity of autocorrelation at a scattering angle (θ) of 173° with Nano ZS.
- d) The correlation function would be accepted when the difference between the measured and the calculated baselines was less than 0.1 %. The mean diameter will be obtained by the Stokes-Einstein relationship. Zeta potential of chitosan/insulin complexes determined by Nano ZS.
- e) The Particle size and Zeta potential were measured in 5 different trials for all the samples.

7. Insulin encapsulation efficiency of the self-assembled nanoparticles

- a) Equal volume of both chitosan and insulin solution were mixed at different weight ratios and immediately vortexed for 1 min with the help of a Vortex mixer.
- b) Then, the mixture was incubated at room temperature for 30 min to completely form the self-assembled complexes. Nanoparticles were then centrifuged at 14,000 rpm/min for 30 min at room temperature. The clear supernatant was analyzed for insulin content using UV-vis spectrophotometer at 280 nm.
- c) The encapsulation efficiency (EE) will be calculated by the following formula:

$$EE (\%) = \frac{Ti - Fi}{Ti} \times 100$$

Where EE (%) is the Encapsulation efficiency in percentage.

Ti = Total amount of insulin.

Fi = Free insulin in the supernatant.

8. Measurement of In-vitro insulin release profile

- a) For the In-Vitro analysis, Different pH levels corresponding to the GI tract of human beings are to be simulated.
- b) To simulate the GI tract pH levels, prepare 3 different buffer solutions with 3 different pH levels (pH 1.2, pH 6.8 and pH 7.4).
- c) The pH 1.2 represents the pH of the stomach which is very acidic, pH 6.8 represents the pH of the Duodenum and pH 7.4 represents the pH level of the Intestine.
- d) To determine the insulin release profiles of the Chitosan-insulin self-assembled nanoparticles, the different test samples need to be immersed in the 3 buffer solutions (At different pH corresponding to GI tract i.e., pH 1.2, pH 6.8 and pH 7.4) with mild agitation.
- e) At specific time interval of 30 minutes, draw a sample from each buffer.
- f) Measure the concentration of the released insulin sample drawn using UV spectrophotometer at 280nm. Follow this step for each sample.

Risk/Safety Discussion

In this project, all materials used are readily available. The experiments were carried out safely. No culture of any kind was done. Food grade Chitosan and readily available lab-grade Insulin will be used. Safety precautions and procedures followed by laboratory tests are followed. During sample preparation and tests, lab coat, disposable gloves and protecting glass for eyes were used by the student. All tests were monitored by the adult supervisor. Care was taken for sterilization of all the equipment, flasks etc. All lab equipment, flask, etc. will be cleaned with 10% bleach solution for min. 10 minutes.

The environmental, health and safety regulation as prescribed by local regulatory agencies were followed for disposal.

Data Table(s) and graph(s):

Data Table(s):

3 Sample Weight ratios for Chitosan(CS)-Insulin(INS) in self-assembled Nanoparticles.

Table-1. Different weight ratios for Chitosan(CS)-Insulin(INS) for self-assembled nanoparticles formation

Sample No.	Molecular Weight (kDa)	Degree of Deacetylation (%)	CS:INS Weight Ratio
1	222.0	80.0	1:1
2	222.0	80.0	2:1
3	222.0	80.0	3:1

Particle Sizes of the self-assembled Chitosan(CS)-Insulin(INS) Nanoparticles

Table-2. Particle sizes of chitosan-insulin nanoparticles (In nm)

Trials	Sample-1 (CS:INS 1:1)	Sample-2 (CS:INS 2:1)	Sample-3 (CS:INS 3:1)
1	205.1	460.3	530.6
2	212.6	464.8	522.4
3	208.3	457.4	542.1
4	210.2	470.6	526.6
5	215.0	465.1	535.0
Mean	210.2	463.6	531.3

Zeta-Potential of Insulin control solution and self-assembled Chitosan(CS)-Insulin(INS) Nanoparticles

Table-3. Zeta Potential of chitosan-insulin nanoparticles (In mV)

Trials	Control Solution (Insulin)	Sample-1 (CS:INS 1:1)	Sample-2 (CS:INS 2:1)	Sample-3 (CS:INS 3:1)
1	-10.5	2.8	10.6	20.5
2	-8.6	3.6	14.3	24.3
3	-6.9	4.0	11.2	18.5
4	-12.1	3.1	15.0	19.6
5	-10.9	2.5	10.9	22.7
Mean	-9.8	3.2	12.4	21.1

Encapsulation Efficiency (%) of self-assembled Chitosan(CS)-Insulin(INS) Nanoparticles

The encapsulation efficiency (EE) will be calculated by the following formula:

$$EE (\%) = \frac{Ti - Fi}{Ti} \times 100$$

Where EE (%) is the Encapsulation efficiency in percentage.

Ti = Total amount of insulin. Fi = Free insulin in the supernatant.

Table-4. Insulin encapsulation (%) in chitosan-insulin nanoparticles.

Trials	Sample-1 (CS:INS 1:1)	Sample-2 (CS:INS 2:1)	Sample-3 (CS:INS 3:1)
1	75.6	82.5	88.5
2	80.2	88.7	90.4
3	82.5	84.2	94.5
4	78.3	80.7	92.6
5	79.4	86.5	95.3
Mean	79.2	84.5	92.3

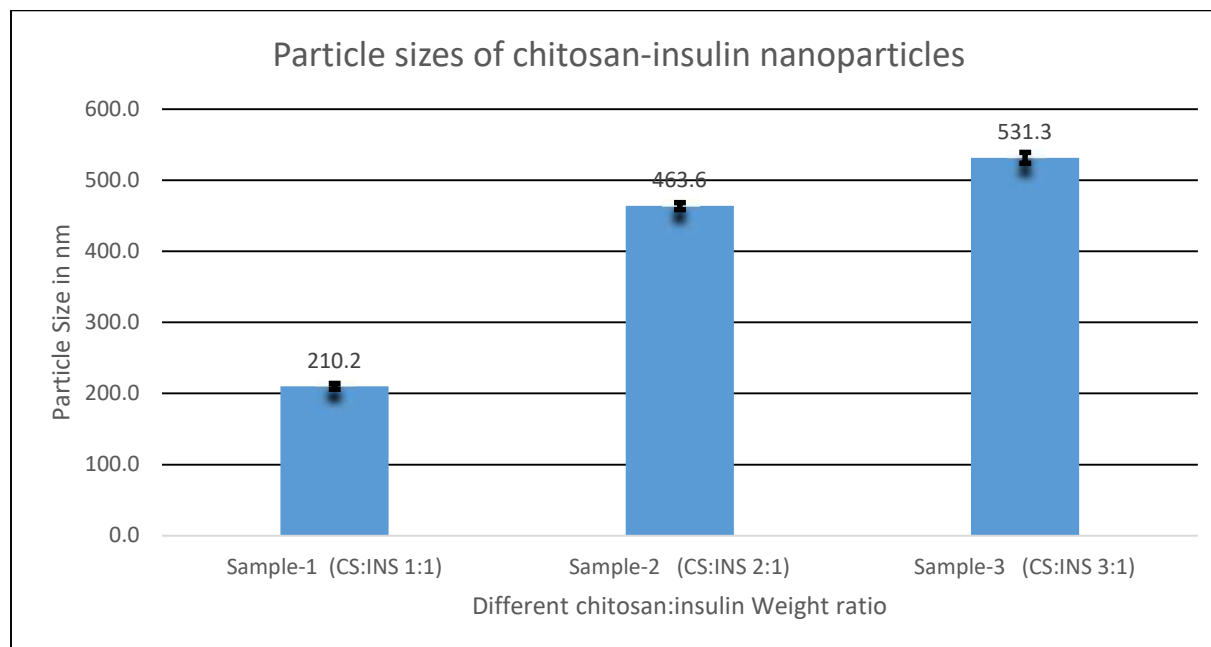
Insulin release profile of self-assembled Chitosan(CS)-Insulin(INS) Nanoparticles

Table-5. Insulin release profile of the chitosan-insulin nanoparticles.

Minutes	Stomach(pH 1.2)	Duodenum (pH 6.8)	Ileum (pH 7.4)
0	0	0	0
30	5.7	28.4	35.5
60	11.5	52.6	60.1
90	12.4	54.7	65.2
120	13.0	55.8	72.4
150	13.8	57.8	75.6
180	14.8	58.9	78.5
210	15.2	62.5	82.4
240	15.4	65.8	85.6

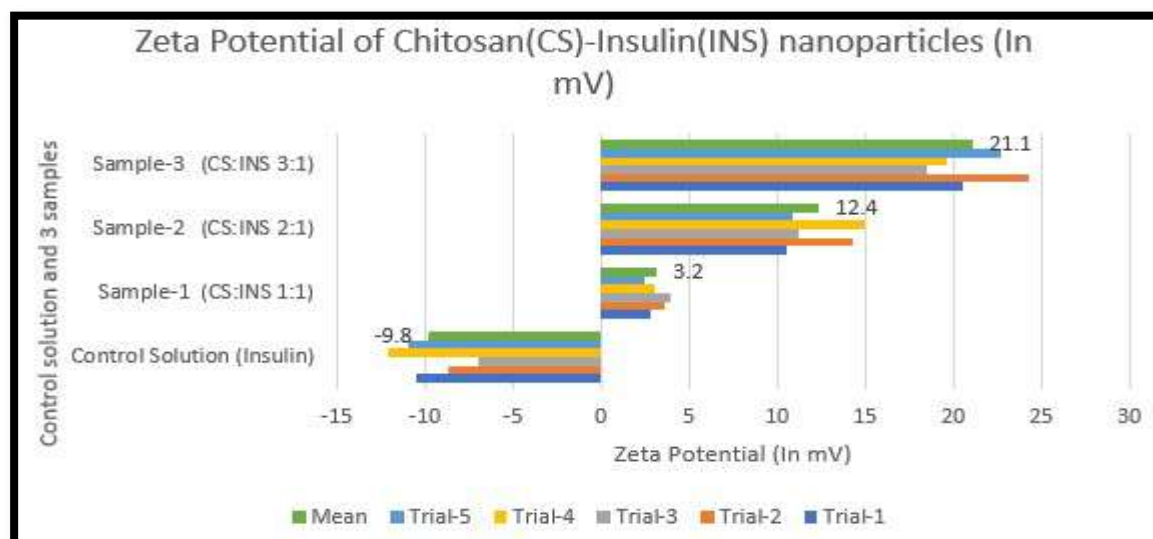
Graph(s):

Particle Sizes of the self-assembled Chitosan(CS)-Insulin(INS) Nanoparticles



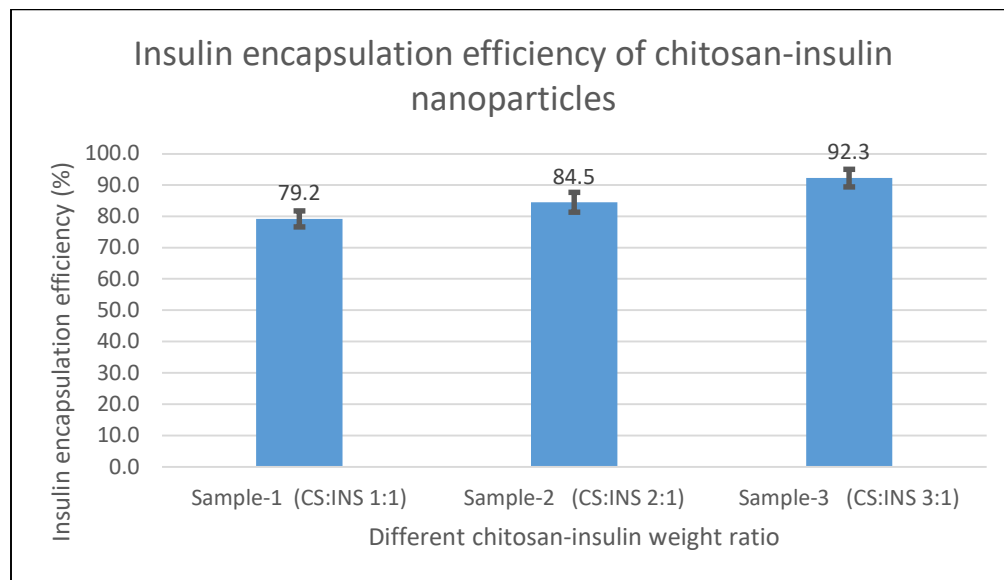
Graph-1. Particle sizes of chitosan-insulin nanoparticles (In nm)

Zeta-Potential of Insulin control solution and self-assembled Chitosan(CS)-Insulin(INS) Nanoparticles



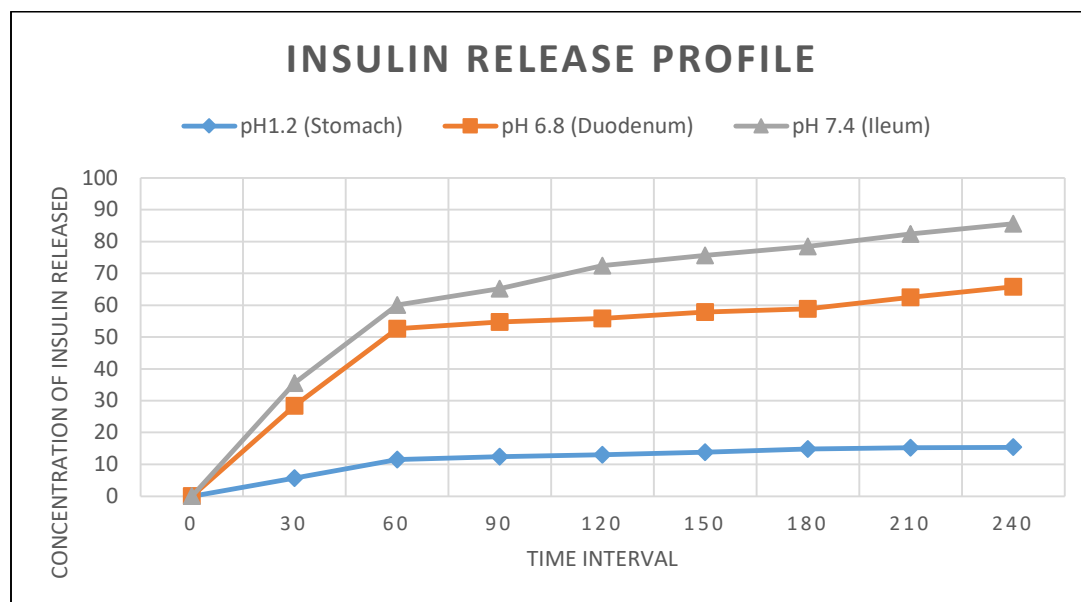
Graph-2. Zeta Potential of chitosan-insulin nanoparticles (In mV)

Insulin Encapsulation Efficiency (%) of self-assembled Chitosan(CS)-Insulin(INS) Nanoparticles



Graph-3. Insulin Encapsulation Efficiency (%)

Insulin release profile of self-assembled Chitosan(CS)-Insulin(INS) Nanoparticles



Graph-4. Insulin Release Profile at different pH corresponding to the GI tract.

Data Analysis

The data analysis of this project consists of 2 main parts.

- a) Determination of the physicochemical properties of the self-assembled nanoparticles, and
- b) In-Vitro Insulin release profile and the Insulin encapsulation efficiency of the self-assembled nanoparticles.

1. Determination of the physicochemical properties of the Self-Assembled nano particles

a) Fourier transform infrared (FT-IR) spectroscopy analysis

The FT-IR spectroscopy graphs of the chitosan, insulin and chitosan-insulin self assembled nanoparticles were captured and analyzed for the intensity peaks.

For chitosan, the intensity peaks at 3427.73 cm⁻¹ (O-H stretch and N-H stretch, overlap), 2922.49 and 2859.82 cm⁻¹ (C-H stretch), 1652.90 cm⁻¹ (NH₂ deformation), 1154.09 cm⁻¹ (bridge-O- stretch) and 1092.74 cm⁻¹ (C-O stretch).

Insulin shows an intensity peak at 1656.33 cm⁻¹ (C=O stretching of amide I) which corresponds to the large α -helical of native insulin and a medium intensity peak at 1537.81 cm⁻¹ of the insulin was attributed to amide-II corresponding C–N stretching and N–H bending modes.

In case of the chitosan-insulin self-assembled nanoparticles, a little shift of amide I region towards lower spectra from 1656.33 to 1641.13cm⁻¹ was found in self assembled nanoparticles compared to that of native insulin. This fact may be attributed to the electrostatic interaction between the negative charges of insulin and positive charges of chitosan.

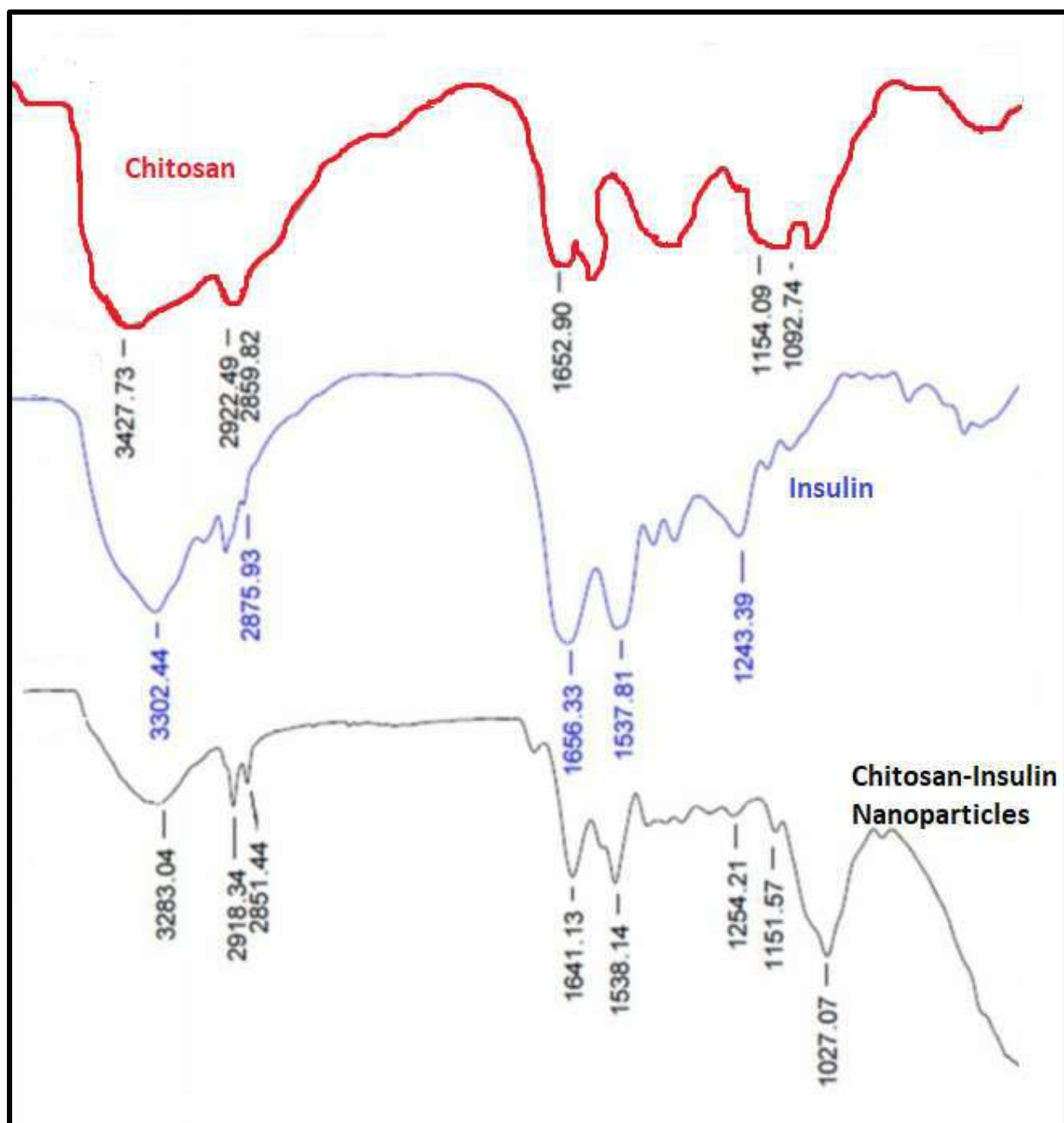


Figure-11. FT-IR Spectrum of Chitosan, Insulin and Chitosan-Insulin nanoparticles.

b) Analysis of the Particle size and zeta potential

The particle size of self-assembled chitosan-insulin nanoparticles was measured by dynamic light scattering (DLS). 5 different trials were carried out measuring the particle sizes of the 3 samples in consideration. The average size of the 5 trials different sample of the nanoparticles was obtained as shown in Table 2. The average particle size of the

Chitosan-insulin weight ratio 1:1 was 210.2 nm. For the sample with weight ratio 2:1, the average size was 463.6, and for the sample with weight ratio 3:1, the particle size was 531.3. This shows that chitosan can condense insulin to form nanoparticles at different weight ratio, the nanoparticles formed at 1:1 weight ratio achieved the lowest particle size and being the smallest, it is preferable for delivering insulin orally.

The zeta potential of the nanoparticle samples and the Insulin control solution were also measured in 5 different trials as shown in Table 3. It was found that the average zeta potential of the Insulin control solution was -9.8 mV. However, the zeta potential of the chitosan-insulin nanoparticles was subsequently higher at +3.2 mv, +12.4 mV and +21.1 mV. Hence the zeta potential increased with the weight ratio. This shows that the positive charges of the chitosan molecules neutralized all the negative charges of the insulin molecules, and achieved positive charges. Also, with the increase of weight ratio, the zeta potential increased due to accumulation of excessive positive charges from chitosan.

2. Insulin encapsulation efficiency of the self-assembled nanoparticles and In-Vitro Insulin release profile in the pH conditions corresponding to the Human GI tract.

a) Insulin encapsulation efficiency

For measuring the insulin encapsulation efficiency (EE), the insulin content in the 3 sample nanoparticles were measured in 5 separate trials using UV-vis spectrophotometer at 280 nm.

The encapsulation efficiency (EE) will be calculated by the following formula:

$$EE (\%) = \frac{Ti - Fi}{Ti} \times 100$$

Where EE (%) is the Encapsulation efficiency in percentage,

Ti = Total amount of insulin. And Fi = Free insulin in the supernatant.

It was found that sample 1 with CS:INS weight ratio achieved the encapsulation efficiency of 79.2%, whereas the sample 2 with 2:1 weight ratio achieved 84.5% and sample 3 with 3:1 weight ratio achieved 92.3% as shown in Table 3. Although sample 2 and sample 3

showed better insulin encapsulation in comparison to sample 1, still sample 1 is further used for in vitro analysis as it showed smallest particle size in DLS analysis. For successful oral insulin delivery, ideally the nanoparticles must have high insulin encapsulation efficiency. Poor encapsulation efficiency might hinder the use of polymeric delivery devices in insulin delivery as it would cause wastage of expensive drug molecules. The result here showed that all the nanoparticles achieved high insulin encapsulation efficiency suitable for the oral insulin delivery.

b) In vitro insulin release profile at different pH values

The Insulin release profile from Chitosan-Insulin self-assembled nanoparticles were performed in the gastric (stomach) and intestinal (duodenum, ileum) simulated pH conditions using the sample with chitosan-insulin ratio of 1:1. The insulin release from nanoparticles both under gastric condition and intestinal condition were measured at 30 minutes intervals for 4 hours. Data was collected for all the measurements and for all the trials to observe the concentration of the released insulin over time in the different pH conditions. As shown in Table 4, It was observed that at the acidic condition of stomach (at pH 1.2), immediate release of insulin from the nanoparticles was low, releasing only 15% of encapsulated insulin. This may be due to the strong ionic interaction between negative charges of insulin and positive charges of chitosan. As chitosan has a pKa value of 6.5, almost entire (> 95%) amino groups of chitosan are protonated in lower pH. Thus, entrapped insulin would be protected from the proteolytic enzymatic degradation in the stomach.

However, in the intestinal conditions of duodenum and ileum, i.e., at pH 6.8 and 7.4 respectively, it achieved significant insulin release from the nanoparticles. In the buffer pH 6.8 (duodenum), the amount of released insulin was almost 65% of the encapsulated amount in 4 hrs. Also, in the initial 1 hr., the insulin release was about 35-40% but over 4 hrs., it was about 65% of encapsulated amount. This prolong release profile will help to produce desired effects within animal system. Similarly, at pH 7.4 mainly weak affinities between ionic groups of chitosan and insulin influenced prolonged and significant insulin release of almost 85%.

Therefore, the in vitro release profile of self-assembled nanoparticles clearly showed a large amount of insulin retention in acidic pH, whereas, a significant amount of insulin released was achieved in higher pH for sustained period of time. This pH responsive release pattern would facilitate oral administration of insulin.

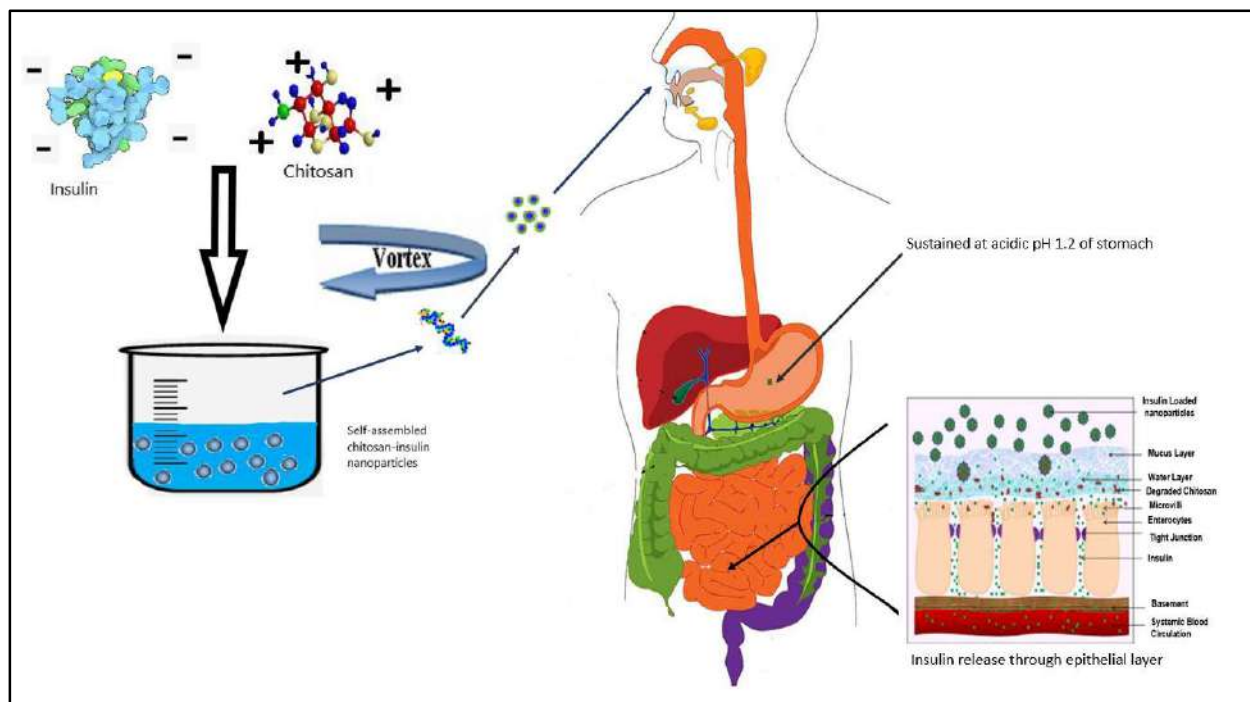


Figure-12. Oral delivery of Insulin using the Chitosan-Insulin nanoparticles.

Conclusion

The project was in 3 parts, the first part was about preparing the self-assembled chitosan-insulin nanoparticles that can be used for oral insulin delivery, the second part was about determining its physicochemical properties and the third part was to determine the suitability for oral delivery of Insulin. The hypothesis was that if 3 different samples were taken with the chitosan-insulin weight ratios of 1:1, 2:1 and 3:1, all the samples will be able to achieve the desired insulin encapsulation efficiency and release profile, and sample with weight ratio 1:1 will be most suitable due to its smallest particle size. The observations and data analysis above clearly demonstrate that the hypothesis is fully accepted. Weight ratio 1:1 obtained average 210.2 nm particle size which is the smallest

among 3 samples. Also, all the samples obtained 79-93% insulin encapsulation efficiency, and in the insulin release profile, it was observed that the nanoparticle was able to achieve large insulin retention in the acidic pH condition of stomach and significant amount insulin was released (up to 85%) in the intestinal pH condition. It also showed that the release was prolonged for over 4 hrs., with 35-40% release in 1 hr., and maximum release in 4 hrs. Hence this project concludes that the nanoparticles prepared by the self-assembling process demonstrated here without using harsh organic chemicals might be employed to improve the potential delivery approach for oral insulin, and chitosan-insulin self assembled nanoparticles can be potentially used for oral insulin delivery.

Overall, this was a fun project to execute, and I learnt what are the different ways of drug delivery and what are the challenges associated with them. I also learned the importance of nanotechnology and how that can be used to solve in some of the challenges.

Future Goal:

In this project I focused on the preparation of the self-assembled nanoparticles without using any harsh chemicals, and also on the determination of the physicochemical properties and in-vitro insulin release profile. However, for determining suitability in oral insulin delivery, many other analyses would be necessary which I could not do due to constraints in the laboratory and approval processes.

- a. In-Vivo analysis using real animals, and study the effect of the nanoparticles on them, and determine the in-vivo insulin release profile.
- b. Toxicological analysis of the prepared nanoparticles, and determining of any harmful effect and how to restrict those.
- c. Effect of external stimulants, temperature, pressure etc. and how those affects the nanoparticles.
- d. Determination of optimum dosage and its effect on reducing blood glucose levels

over time.

- e. Integrating bioremediation process with the existing wastewater treatment plants assess its performance in more realistic environment.

Therefore, the future goal of my project is to focus on these aspects of oral insulin delivery using the self-assembled nanoparticles.

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Noninvasive treatment of Diabetes

An In-Vitro Analysis of Chitosan-Insulin Nanoparticles for Oral Insulin Delivery.

Hrishika Roychoudhury

Grade 11

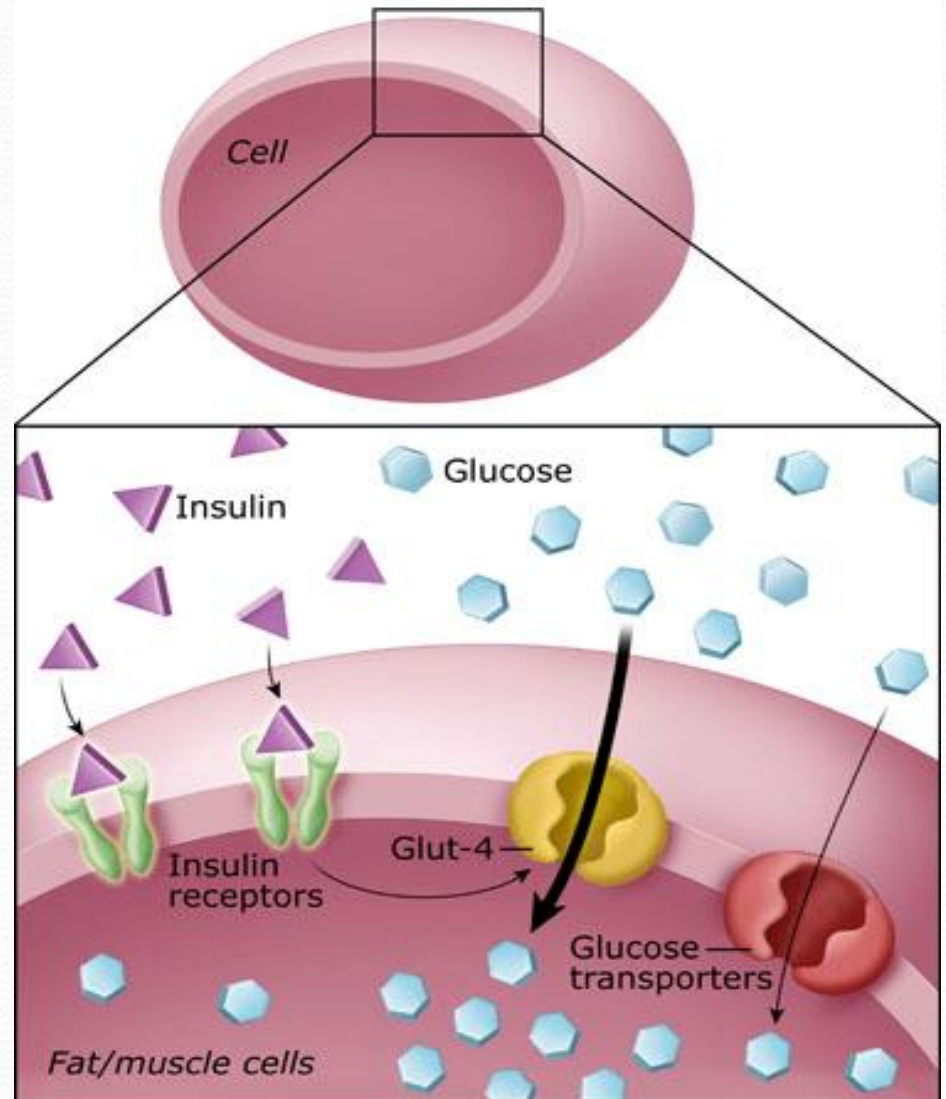
North Carolina School of Science of Mathematics



What is Insulin?

- ❖ **Insulin is a protein hormone secreted by pancreatic β -cells.**
- ❖ **Plays a crucial role in glucose metabolism regulation**

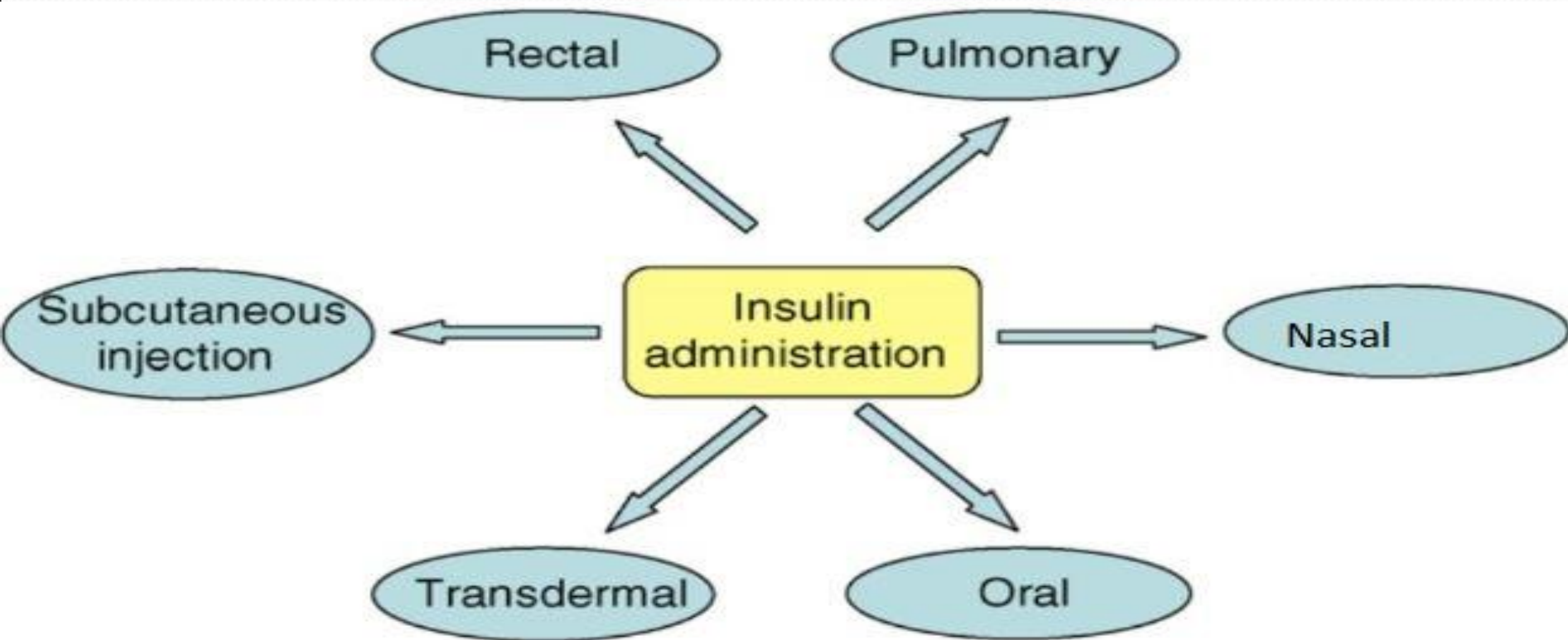
The Effect of Insulin



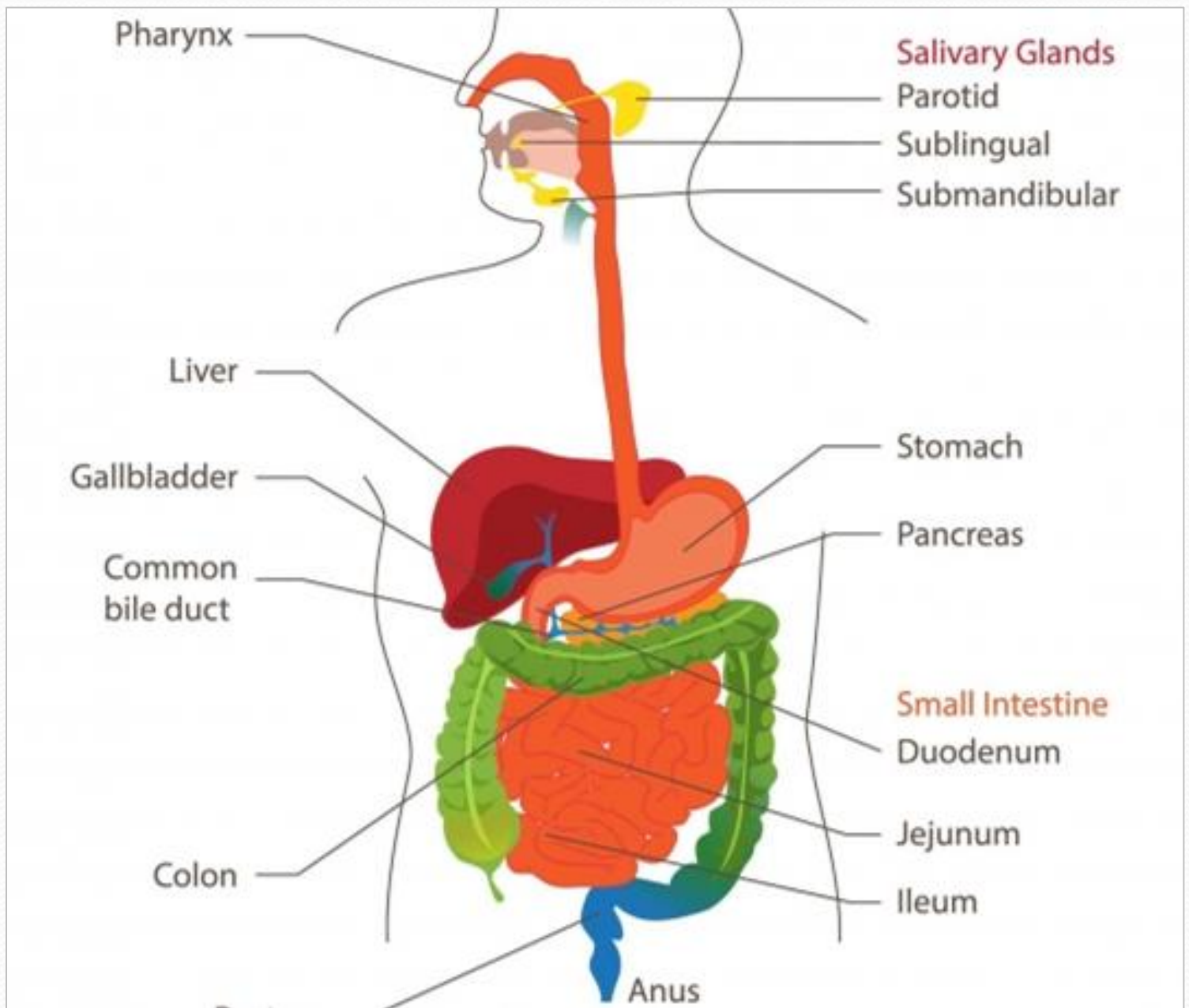
Diabetes

- ❖ Diabetes mellitus is a metabolic disease characterized by high blood sugar levels due to defects in insulin secretion or the body's ability to use insulin.
 - **Type 1 diabetes (5-10% of diabetes patients)**
 - No Insulin due to damage in the pancreas β -cells.
 - **Type 2 diabetes (90-95% of diabetes patients)**
 - Body does not effectively use insulin.
 - **Other types of diabetes**
 - **Gestational diabetes** - occurs from pregnancies.

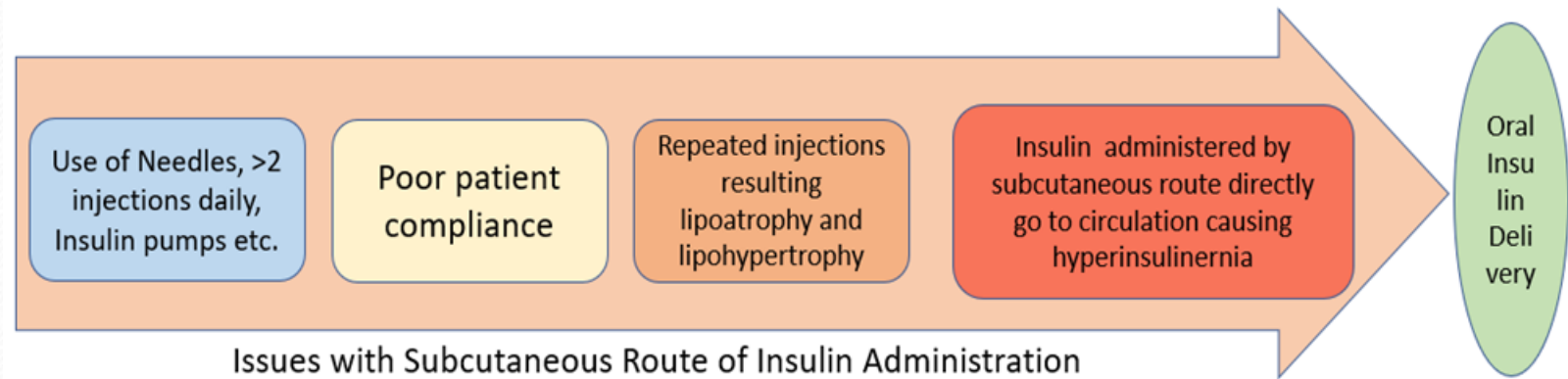
Different routes of Insulin delivery



GI Tract

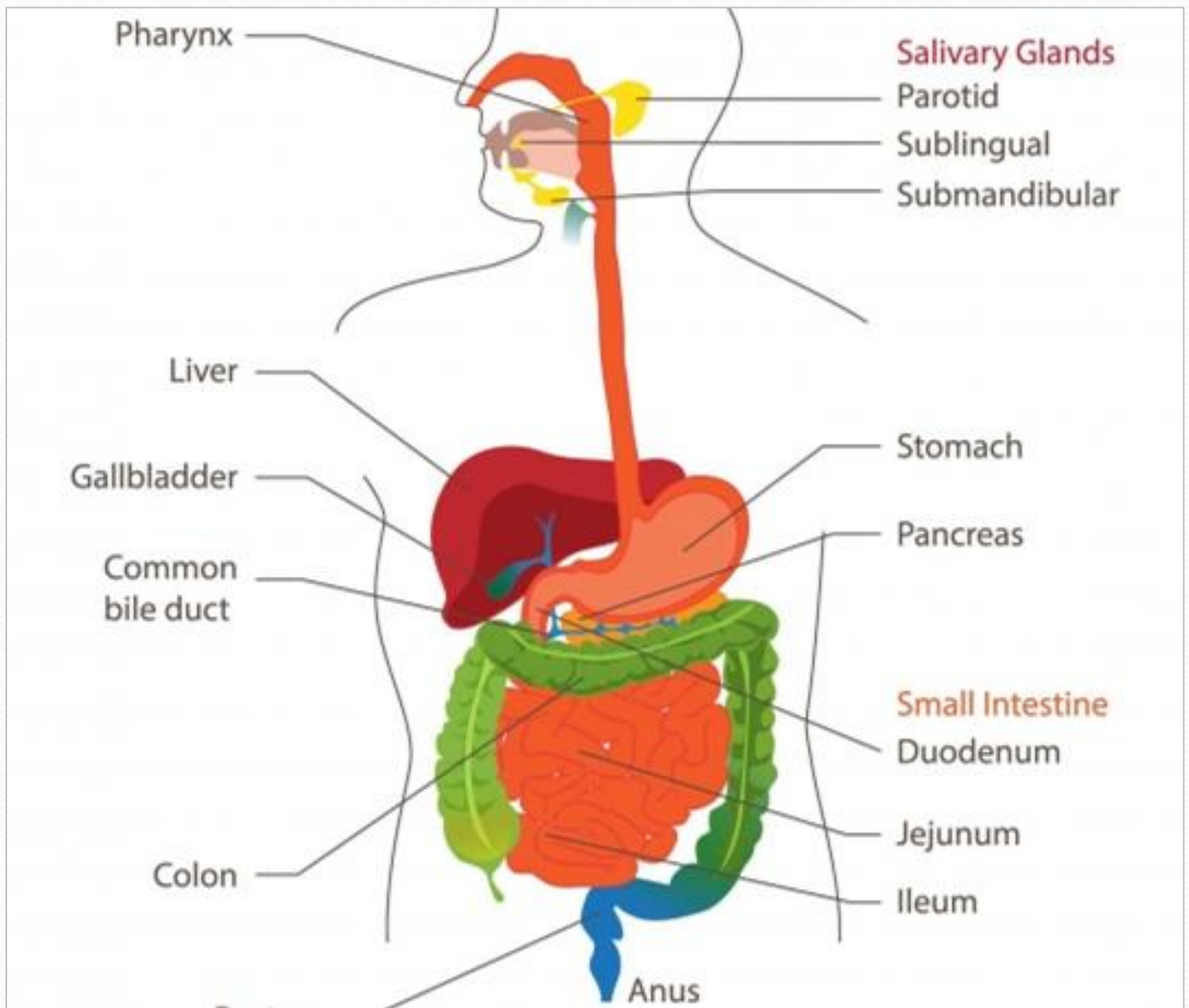


Oral Insulin Delivery

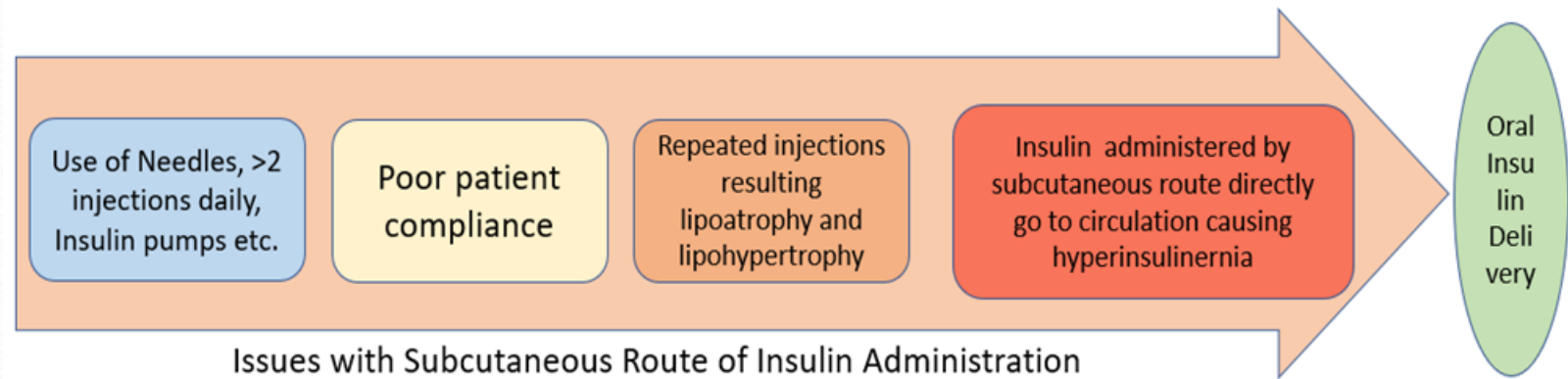


- ❖ **Oral delivery:** preferred route of drug administration.
- ❖ **Better patient compliance due to noninvasive nature.**
- ❖ **Cost-effectiveness**
- ❖ **Less pain and immune reactions.**
- ❖ **Better mimics the normal insulin pathway.**
- ❖ **High portal vein concentrations, with no sustained peripheral hyperinsulinemia.**

GI Tract



Oral Insulin Delivery



- ❖ **Oral delivery:** preferred route of drug administration.
- ❖ **Better patient compliance due to noninvasive nature.**
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- ❖ **Better mimics the normal insulin pathway.**
- ❖ **High portal vein concentrations, with no sustained peripheral hyperinsulinemia.**

Limitation of Oral insulin delivery

Physical Barriers

- Mucous Layer
- Water Layer
- Intestinal Epithelium
- Tight Junctions

Biochemical Barriers

- Acidic pH conditions
- Enzymatic degradation

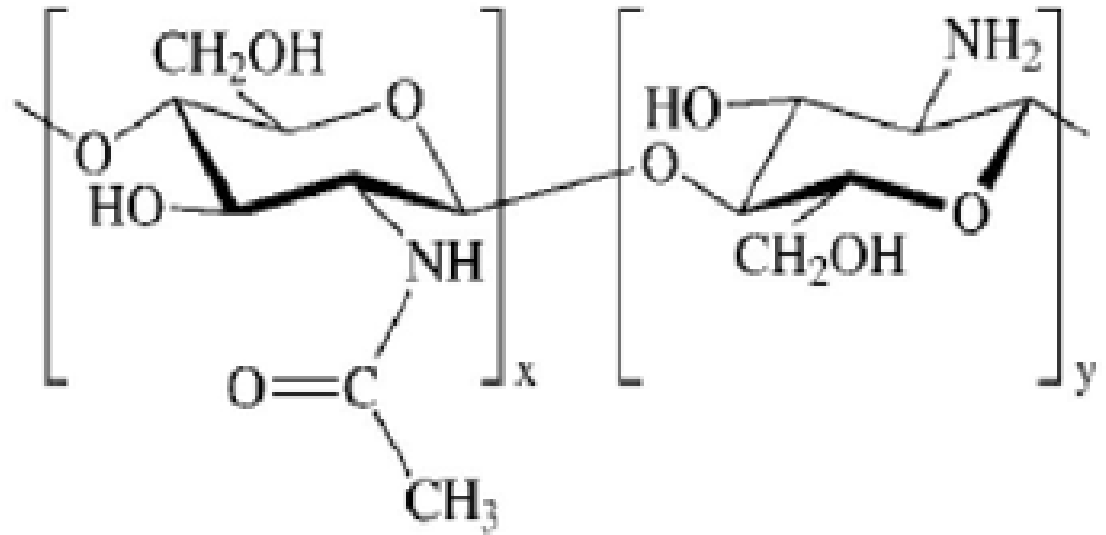
Use of Nanotechnology in drug delivery

- Recent advances in Nanotechnology have made the mass production of therapeutic micro-molecules, such as proteins, nucleic acids and polysaccharides, feasible.
- The nanoparticles can help in drug delivery due to its versatility in targeting tissues, enabling deep molecular targets as well as sustained drug release.
- Nanoparticles display excellent efficiency as potent proteins, peptide, vaccines and DNA carrier.
- Thus, nanoparticles offer a better way for delivering the hydrophilic drugs such as insulin.
- Several **natural polymers like, alginate, pectin, gelatin, chitosan, etc.** can be used to prepare the nanoparticles.

Why Chitosan

- ❖ **Biodegradable**
- ❖ **Biocompatible**
- ❖ **Non-toxic**
- ❖ **Muco-adhesive**
the GI tract.

- ❖ **Cationic due to protonation of amine groups** - giving effective encapsulation of bio-molecules like proteins, drugs and binding effects to epithelial cells.



Chitosan extraction

Crabs
Shrimps

Chemical extraction



Biological extraction

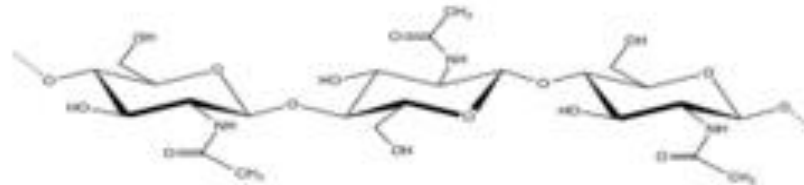
Dilute aqueous HCL solution

Decalcification

Deproteination

Proteases producing bacteria

Decolorization



Deacetylation

Hot concentrated
NaOH solution
(40-50%)

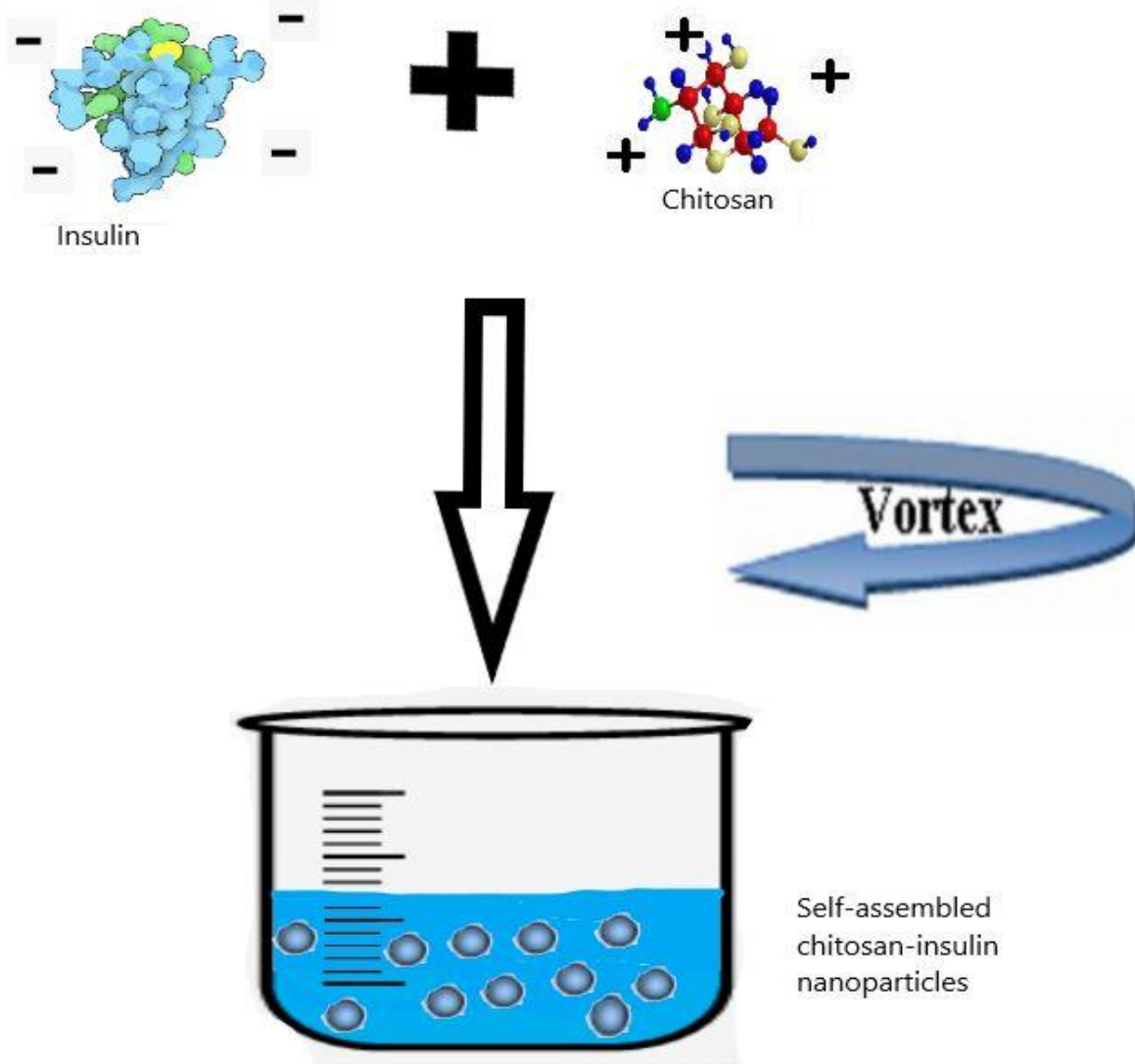
Chitin deacetylase



Chitin

Chitosan

Self-assembly of Chitosan-Insulin nanoparticles



- ✓ Formation of polyelectrolyte-complexes by electrostatic interaction. Optically homogeneous, stable nano-dispersion at colloidal level.
- ✓ The interactions are **Vander Waals force, hydrogen bonding, coulombic forces or transfer forces.**

Objectives of the Project

- **Make a self-assembly of Chitosan and Insulin nanoparticles** by encapsulating insulin into chitosan without using any harsh chemicals.
- **Determination of the physicochemical properties** of the Chitosan-Insulin nanoparticles.
- **In-Vitro analysis of the nanoparticles for Insulin encapsulation efficiency and Insulin release profile in Human digestive system simulated pH conditions.**



Question:

How do different weight ratios of Chitosan-Insulin nanoparticles improve insulin retention and release at different pH corresponding to the human digestive system GI tract?

Hypothesis:

If 3 samples of different weight ratios (1:1, 2:1 and 3:1) of Chitosan and Insulin are taken in the self-assembled Chitosan-Insulin nanoparticle, their particle sizes are analyzed and they are put in the pH conditions corresponding to the human digestive system, then the **Chitosan-Insulin weight ratio 1:1 will be able to achieve the lowest particle size and all the weight ratios of the self-assembled nanoparticles will be able to achieve more concentration of released Insulin than the control sample of only insulin** without any Chitosan nanoparticle carrier.

Independent Variables: 3 different weight ratios of Chitosan - Insulin nanoparticles.

Dependent Variables: Particle Size, Zeta potential and Concentration of Insulin.

Constants: The pH conditions, Volume of the samples, Degree of deacetylation of the samples, Temperature, etc.

Materials:

- ❖ Glacial Acetic Acid. 60.052 g/mol
1.049 g/cm³
- ❖ Tris (hydroxymethyl) Aminomethane
500 g
- ❖ Chitosan 70-150 mPa.s
- ❖ Insulin 27 USP units/mg
- ❖ Distilled Water 10 L
- ❖ Erlenmeyer flasks 500 ml - 8 Nos
- ❖ Graduated cylinder 100 ml - 4 Nos
- ❖ Glass funnels and filter stand with
Burette clamp.
- ❖ Test tubes - 20 nos
- ❖ Hydrochloric Acid - 30 ml
- ❖ Millipore 0.5 µm filter
- ❖ Sodium Hydroxide (NaOH) solution -
100 ml
- ❖ Sodium Nitrite (NaNO₂) - 50 gm
- ❖ Portable pH meter.
- ❖ Low Profile Vortex Mixer.
- ❖ UV-vis Spectrophotometer.
- ❖ Infrared Spectroscope.
- ❖ Zetasizer Nano ZS.
- ❖ Glass Capillary Viscometer.

Procedures:

Chitosan nanoparticle preparation

Main aspects of Chitosan Nanoparticles:

- Two main aspects of chitosan play the most significant role in determining the suitability of chitosan for drug delivery.

Molecular Weight (MW) and Degree of deacetylation (DD).

- These affects the chemical, physical and biological properties such as DNA condensation and release, immune-adjuvant activity and cellular uptake.

Purification of Chitosan:

Chitosan is purified by oxidative degradation with Sodium Nitrate NaNO_2 at room temperature.

- ❖ Make 1% acetic acid by adding 2 ml 99.8% Glacial acetic acid in 198 ml distilled water.
- ❖ Weigh 2 g of dry chitosan and pour 200 ml of the 1% acetic acid solution in it.
- ❖ Stir the solution to dissolve the chitosan to make 1%(w/v) chitosan dissolved in 1% acetic acid.
- ❖ Add 10 ml of NaNO_2 in 1% acetic acid drop wise for 30 min and stir it vigorously.
- ❖ Keep the mixture at room temperature for 3 h for the reaction to complete.
- ❖ Neutralize the reaction mixture with 1N NaOH to pH 7.4 to complete precipitation of chitosan.
- ❖ Wash the precipitate with distilled water, and dry it overnight at 55°C to obtain purified chitosan.

Procedures:

Self-assembly of chitosan-insulin nanoparticles

- ❖ Dissolve the dry purified chitosan made before in 1 % acetic acid solution and stir for 30 min.
- ❖ Adjust the pH of the solution at 5.5 -5.7 by the addition of sodium hydroxide solution.
- ❖ Dissolve lab-grade Insulin (27 USP/ml) in 0.1M HCl solution.
- ❖ Adjust the pH of the Insulin solution to 8.4 by 0.1 N Tris (hydroxymethyl) aminomethane solution.
- ❖ Mix equal volume of both chitosan and insulin solutions and with a Vortex mixer, vortex for 30 sec.
- ❖ Incubate the mixture at room temperature for 30 min to completely form the self-assembled complexes.



Procedures:

● termination of molecular weight (MW) by glass capillary viscometer

The molecular weight is determined by first measuring the viscosity of the solution. Due to the pressure difference at the top and bottom of the column of liquid, the capillary corrections of flow times according to capillary type is introduced in the calculations, using the formula provided by the capillary manufacturer.

Determination of Degree of Deacetylation (DD) by titration

DD (%) of chitosan is calculated by using the following equation:

$$DD (\%) = \frac{203.2}{42.0 + \frac{1000m}{C_{NaOH} (V2 - V1)}} \times 100$$

- **m** is the amount of chitosan (g) in the solution. **C_{NaOH}** is the concentration of NaOH solution (mol/L).
- **V2 – V1** is the volume of NaOH solution consumed in titration of amino groups of chitosan.
- **203.2** is the molecular weight of the acetylated monomeric unit of the polysaccharide.
- **42.0** is the MW difference between the deacetylated monomeric unit and acetylated monomeric unit.
- **1000** is the conversion factor of milliliters to liters and 100 is the conversion factor of DD to percent.

Procedures:

Determination of Physicochemical Properties

Particle Size measurement

- ❖ The particle sizes and distributions of the self-assembled chitosan-insulin nanoparticles were determined by Dynamic light Scattering (DLS) scan.

Zeta Potential measurement

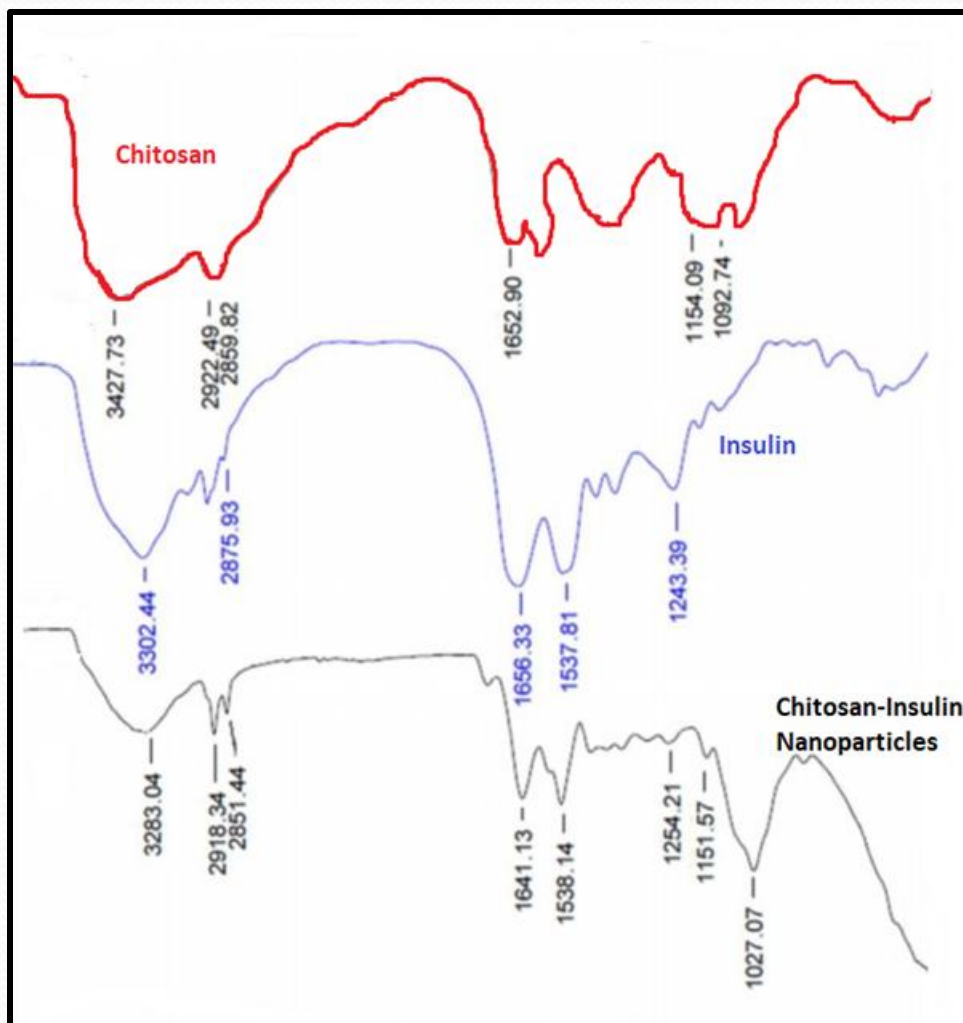
- ❖ Measure the intensity of autocorrelation at a **scattering angle (θ) of 173°** with Nano ZS.
- ❖ **5 different trials** for all the samples.

Fourier Transform Spectroscopy Analysis

- ❖ Fourier transform infrared (FTIR) analysis is done using ATR-FTIR spectrometer.
- ❖ Use the Chitosan solution, Insulin solution and the prepared Chitosan-Insulin nanoparticle solution, and **scan separately from 4000 to 500 cm^{-1} at room temperature.**

Result and Data Analysis

Fourier Transform (FT-IR) Infrared



- ❖ Analyzed for the intensity peaks.
- ❖ Chitosan intensity peaks at 3427.73 cm⁻¹ (O-H stretch and N-H stretch)
- ❖ 2922.49 and 2859.82 cm⁻¹ (C-H stretch).
- ❖ Insulin shows an intensity peak at 1656.33 cm⁻¹ (C=O stretching of amide I) which corresponds to the large α -helical of native insulin.
- ❖ The nanoparticles show little shift of amide I region towards lower spectra from 1656.33 to 1641.13cm⁻¹ due to the electrostatic interaction between

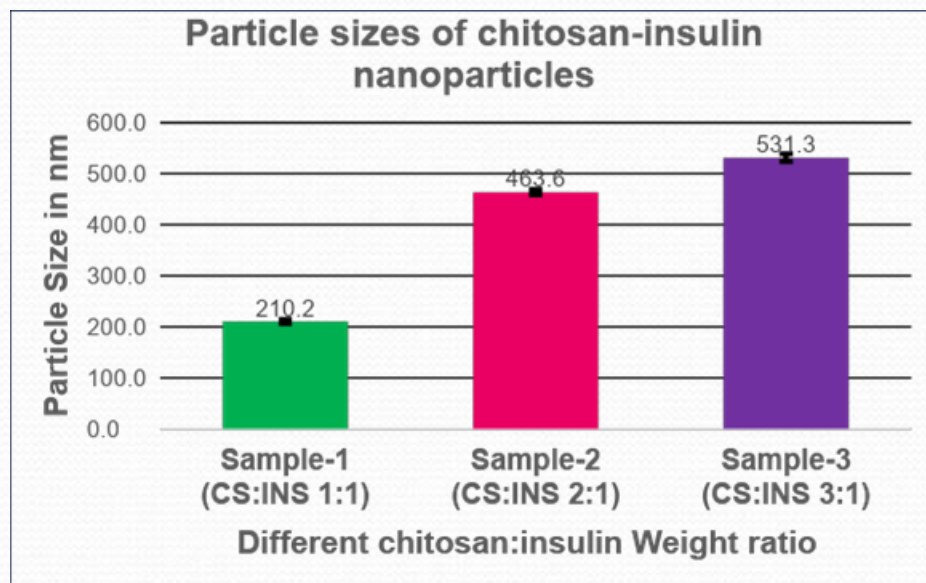
Result and Data Analysis

Molecular Weight and Degree of Deacetylation of 3 samples

Sample No.	Molecular Weight (kDa)	Degree of Deacetylation (%)	CS:INS Weight Ratio
1	222.0	80.0	1:1
2	222.0	80.0	2:1
3	222.0	80.0	3:1

Particle size of Nanoparticles

Trials	Sample-1 (CS:INS 1:1)	Sample-2 (CS:INS 2:1)	Sample-3 (CS:INS 3:1)
1	205.1	460.3	530.6
2	212.6	464.8	522.4
3	208.3	457.4	542.1
4	210.2	470.6	526.6
5	215.0	465.1	535.0
Mean	210.2	463.6	531.3

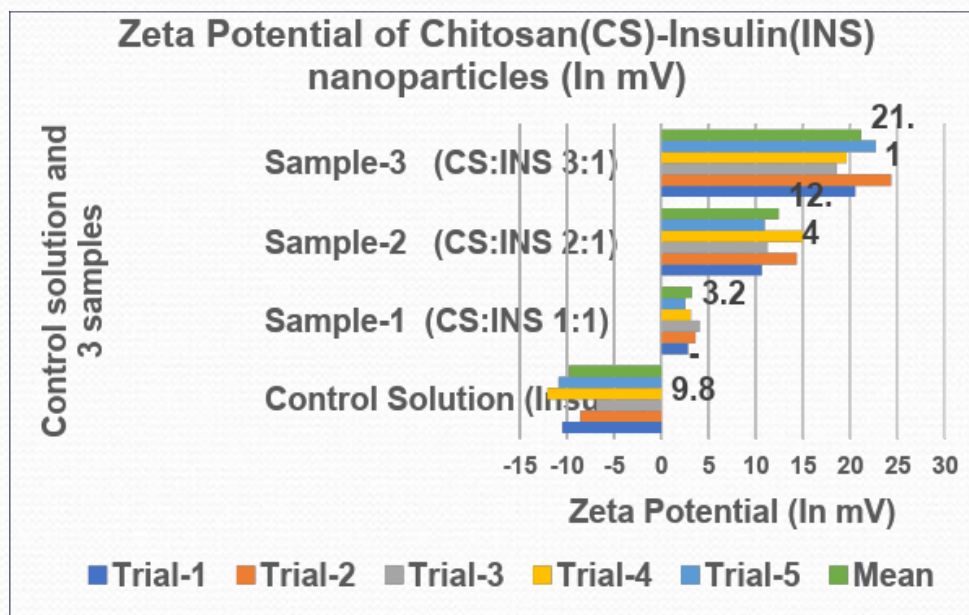


- ❖ This shows that chitosan can condense insulin to form nanoparticles at different weight.
- ❖ Nanoparticles formed at 1:1 weight ratio achieved the lowest particle size.

Result and Data Analysis

Zeta Potential of Insulin solution and the Chitosan-Insulin Nanoparticles

Trials	Control Solution (Insulin)	Sample-1 (CS:INS 1:1)	Sample-2 (CS:INS 2:1)	Sample-3 (CS:INS 3:1)
1	-10.5	2.8	10.6	20.5
2	-8.6	3.6	14.3	24.3
3	-6.9	4.0	11.2	18.5
4	-12.1	3.1	15.0	19.6
5	-10.9	2.5	10.9	22.7
Mean	-9.8	3.2	12.4	21.1



- ❖ Mean zeta potential of the Insulin control solution is -9.8 mV.
- ❖ However, the zeta potential of chitosan-insulin nanoparticles are subsequently higher at +3.2 mV, +12.4 mV and +21.1 mV
- ❖ Positive charges of the chitosan molecules neutralized negative charges of insulin.

Procedures:

Insulin Encapsulation Efficiency

The chitosan-insulin nanoparticles were analyzed for insulin encapsulation efficiency using UV-vis spectrophotometer at 280 nm

The encapsulation efficiency (EE) is calculated by the formula: $EE (\%) = (Ti - Fi) / Ti \times 100$

- Where EE (%) is the Encapsulation efficiency in percentage,
- Ti = Total amount of insulin and Fi = Free insulin

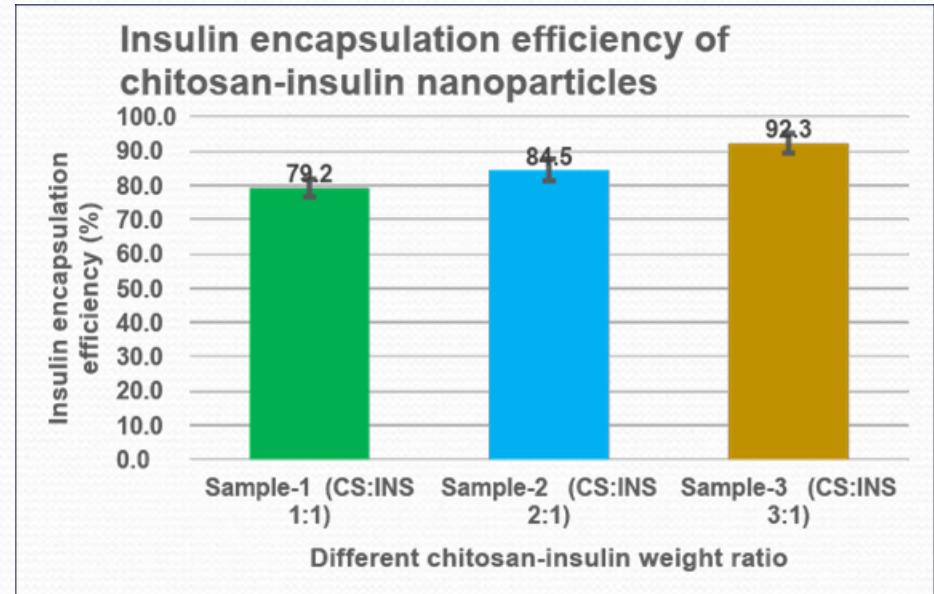
In-Vitro Insulin release profile

- ❖ **Simulated pH levels** corresponding to the **human digestive system** for the In-Vitro analysis.
- ❖ 3 buffer solutions with 3 different pH levels (**pH 1.2, pH 6.8 and pH 7.4**).
 - ❖ **pH 1.2** represents the pH of the **stomach which is very acidic**,
 - ❖ **pH 6.8** represents the pH of the **Duodenum** and
 - ❖ **pH 7.4** represents the pH level of the **Ilium**.
- ❖ Immerse the Chitosan-insulin self-assembled nanoparticles in the 3 buffer solutions with mild agitation.
- ❖ At specific time interval of 30 min, draw a sample from each buffer.

Result and Data Analysis

Insulin Encapsulation Efficiency of the Chitosan-Insulin Nanoparticles

	Sample-1 (CS:INS 1:1)	Sample-2 (CS:INS 2:1)	Sample-3 (CS:INS 3:1)
Trials			
1	75.6	82.5	88.5
2	80.2	88.7	90.4
3	82.5	84.2	94.5
4	78.3	80.7	92.6
5	79.4	86.5	95.3
Mean	79.2	84.5	92.3



$$EE (\%) = (Ti - Fi) / Ti \times 100$$

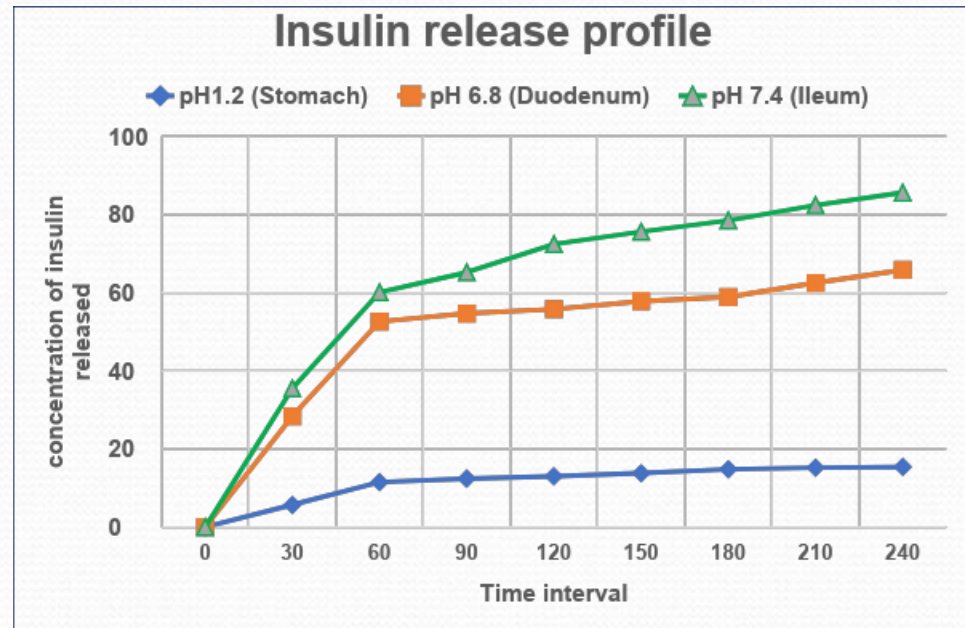
Ti = Total amount of insulin. *Fi* = Free insulin.

- ❖ Sample 1 with weight ratio 1:1 achieved the encapsulation efficiency of 79.2%, weight ratio 2:1 achieved 84.5% and weight ratio 3:1 achieved 92.3%.

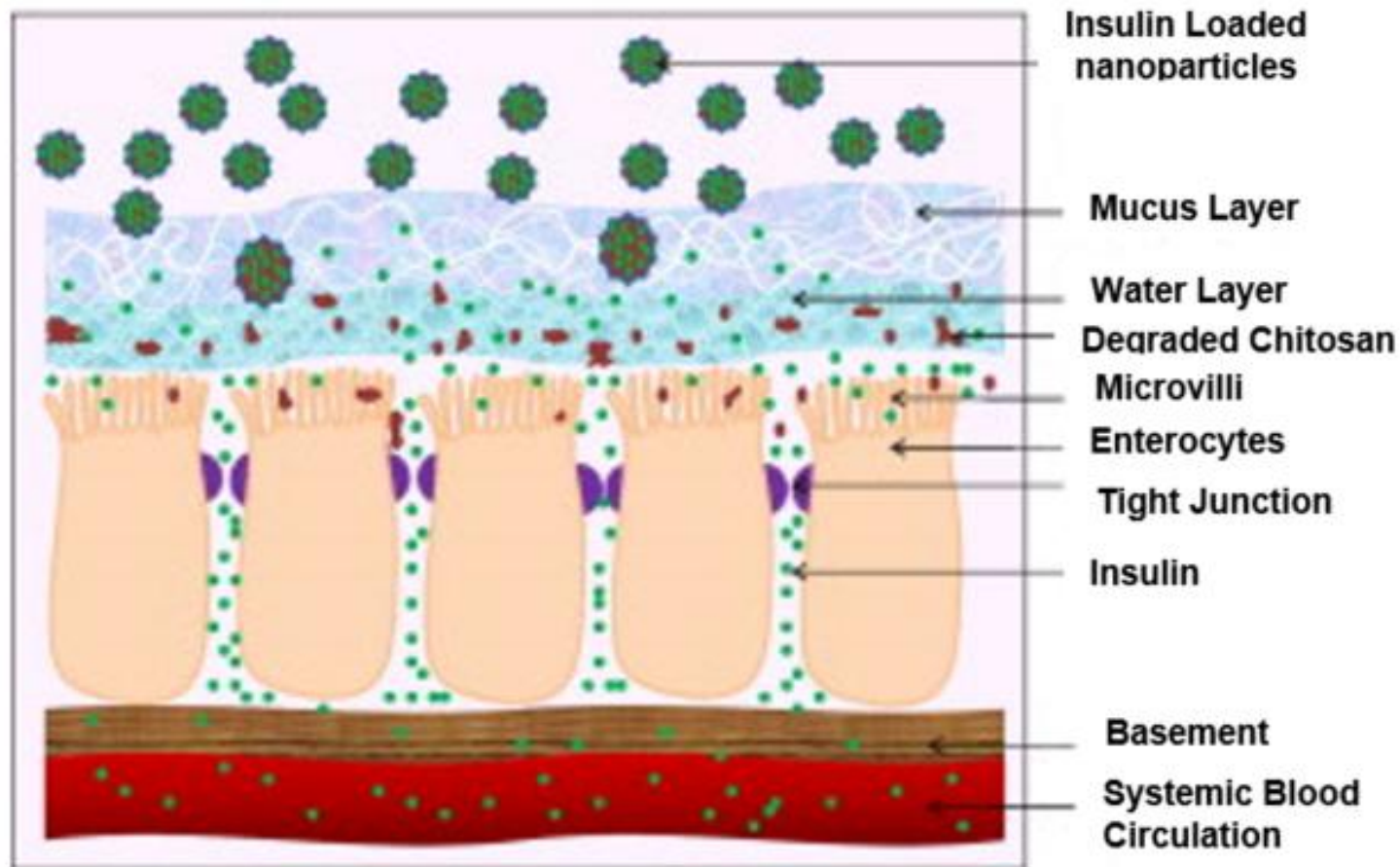
Result and Data Analysis

In-Vitro analysis of Insulin release profile

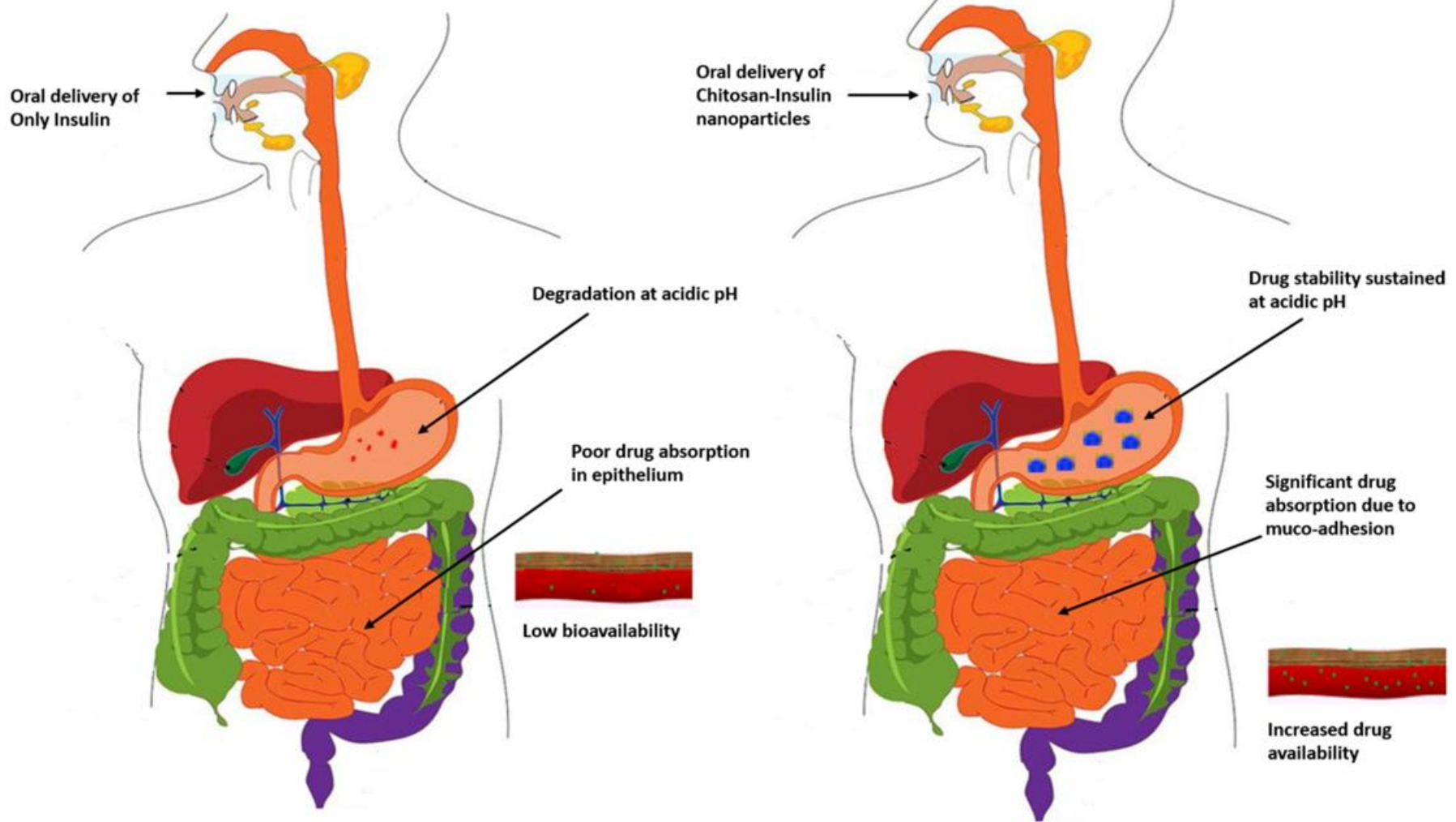
Minutes	Stomach (pH 1.2)	Duodenum (pH 6.8)	Ileum (pH 7.4)
0	0	0	0
30	5.7	28.4	35.5
60	11.5	52.6	60.1
90	12.4	54.7	65.2
120	13.0	55.8	72.4
150	13.8	57.8	75.6
180	14.8	58.9	78.5
210	15.2	62.5	82.4
240	15.4	65.8	85.6



- ❖ At acidic condition of stomach (pH 1.2), immediate release of insulin is **only 15%** of encapsulated insulin due to the strong ionic interaction insulin and chitosan.
- ❖ Almost entire (> 95%) amino groups of chitosan are protonated in lower pH protecting the entrapped insulin from enzymatic degradation in the stomach.
- ❖ In pH 6.8 (duodenum), 65% insulin released in 4 hrs. with 50% in initial 1 hr.
- ❖ This prolong release profile produces desired effects of insulin release over long period.

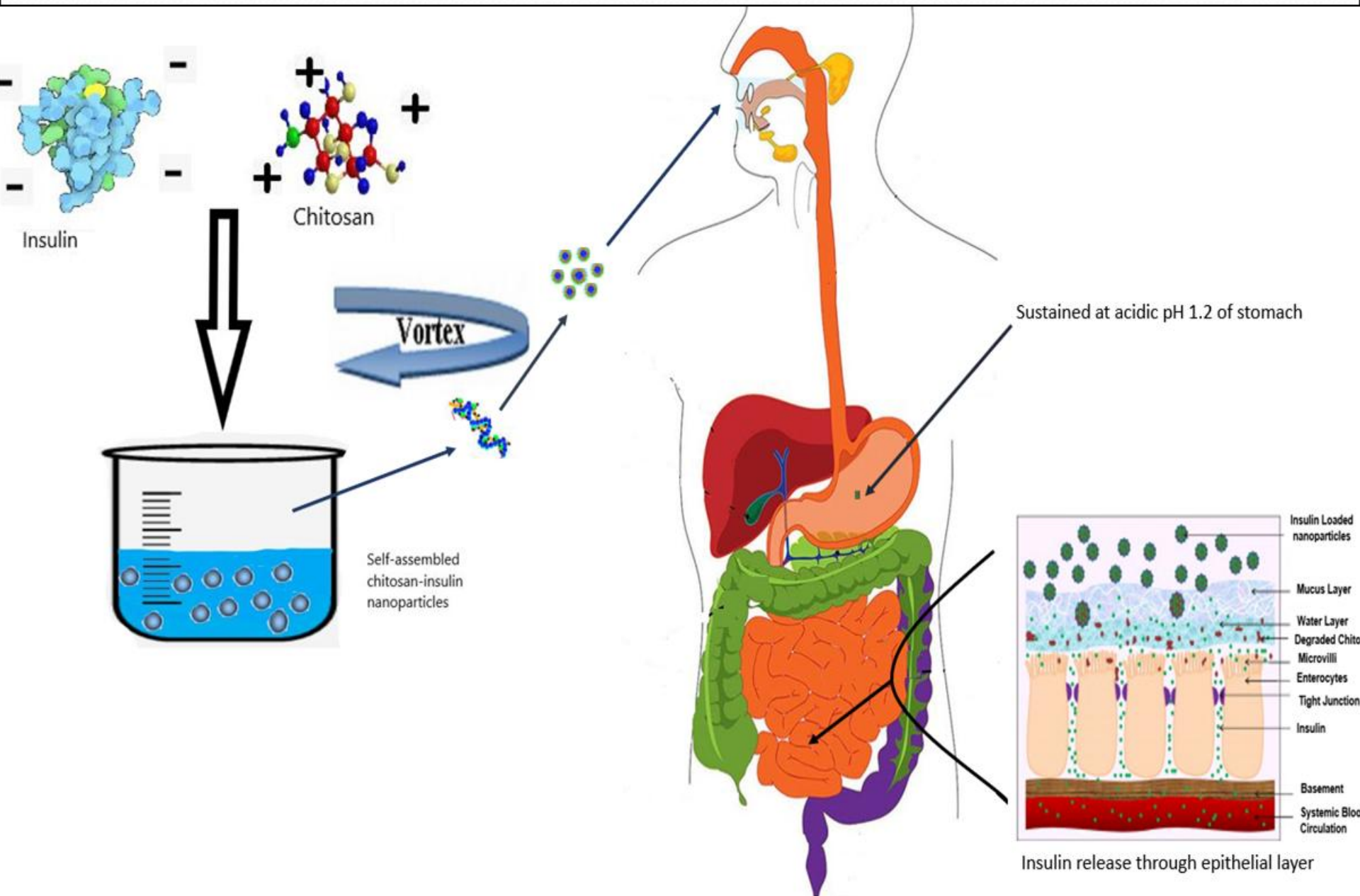


- ❖ The Chitosan-Insulin nanoparticles go through the Mucus layer and water layer.
- ❖ Due to the muco-adhesive nature and positive charges of chitosan, it prolongs the release of insulin that goes through the tight junctions.
- ❖ Smaller particle sizes of the nanoparticles help here.
- ❖ This results the insulin particles reaching the systemic blood circulation and to the



- ❖ Comparing the oral delivery of only Insulin and the Chitosan-Insulin nanoparticles.
- ❖ For only insulin, it encounters degradation in Stomach acidic pH, and poor drug absorption in epithelial layers, resulting low bioavailability.
- ❖ However, The Nanoparticles are sustained in stomach, and achieve significant absorption resulting increased insulin availability in bloodstream.

Graphical Abstract:



Conclusion:

- ❖ The observations and data analysis from the experiment clearly demonstrate that the hypothesis is fully accepted.
- ❖ The nanoparticles were prepared successfully by the self-assembling process without using any harsh organic chemicals that may be harmful for human body.
- ❖ The sample with **weight ratio 1:1 obtained average 210.2 nm particle size** which is the smallest among 3 samples.
- ❖ **All the samples obtained 79-93% insulin encapsulation efficiency.**
- ❖ **At acidic condition of stomach (pH 1.2), release of insulin is only 15%.**
- ❖ Almost entire (> 95%) amino groups of chitosan are protonated in lower pH protecting the entrapped insulin from enzymatic degradation in the stomach.
- ❖ **In pH 6.8 (duodenum), 65% insulin released in 4 hrs. with 50% in initial 1 hr.**
- ❖ This prolonged release produces desired effects of insulin delivery.
- ❖ Similarly, **at pH 7.4, prolonged and significant insulin release of almost 85%.**
- ❖ The project concludes that the nanoparticles prepared by the self-assembling process without using harsh organic chemicals might be employed to improve the potential delivery approach for oral insulin, and chitosan-insulin self assembled nanoparticles

Future Goals:

- ❖ **In-Vivo analysis using real animals**, and study the effect of the nanoparticles on them, and determine the in-vivo insulin release profile.
- ❖ **Toxicological analysis** of the prepared nanoparticles, and determining of any harmful effect and how to restrict those.
- ❖ **Effect of external stimulants**, temperature, pressure etc. and how those affects the nanoparticles.
- ❖ **Determination of optimum dosage** and its effect on reducing blood glucose levels over time.

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Thank You for Your Attention

